Building Segregation: The Long-Run Neighborhood Effects of American Public Housing*

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Abstract

This paper studies the neighborhood effects of mid-century American public housing. I construct a new national dataset containing the locations, completion dates, and characteristics of the near-universe of public housing projects built between 1935 and 1973, which I link to neighborhood-level data from 1930 to 2010. I document that public housing projects were systematically targeted toward poorer neighborhoods with a higher share of Black residents, consistent with the program's slum clearance goals and racialized site selection politics. Using matched difference-in-differences designs, I find that public housing construction caused large, persistent increases in Black population shares and substantial declines in median incomes and rents in recipient neighborhoods. Surrounding neighborhoods within one kilometer also experienced racial transition driven by white population outflows, with limited evidence of economic deterioration. In recipient neighborhoods, the effects are driven particularly by early family projects, opened between 1940 and 1960. Among early projects, white residents left recipient and nearby neighborhoods only around projects whose tenants were or became majority Black. Low-income children who grew up in recipient and nearby neighborhoods experienced lower upward mobility and higher adult incarceration rates, gaps that shrink by more than half once 1980 neighborhood composition is held fixed. These findings demonstrate that mid-century public housing, despite intentions of neighborhood revitalization, reinforced existing patterns of racial and economic segregation.

JEL Codes: H42, J15, J62, R23, R38
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1 Introduction

> The Congress hereby declares that the general welfare and security of the Nation and the health and living standards of its people require housing production and community development sufficient to remedy the housing shortage, eliminate substandard housing through the clearance of slums and blighted areas, and achieve as soon as feasible the goal of a decent home and suitable living environment for every American family.
>
> — *Housing Act of 1949*

Between the 1930s and 1970s, the United States financed the construction of approximately one million federally funded public housing units in one of the most ambitious urban policy initiatives of the 20th century (Shester 2013). Designed to clear slums, address urban housing shortages, and provide affordable housing to low-income families, the American public housing program has become widely synonymous with policy failure. Scholarly and popular accounts have argued that the program created and entrenched racial and economic segregation (e.g., Rothstein 2017; Hirsch 1998; Massey and Denton 1993) and accelerated mid-century urban decline (Jackson 1985). Large megablock projects like Cabrini-Green, Pruitt-Igoe, and the Robert Taylor Homes garnered notoriety and were criticized for concentrating poverty, promoting crime, and destroying the urban fabric of neighborhoods (Jacobs 1961; Newman 1972). Federal policy since the 1970s has reflected this negative view, as the federal government has shifted funding toward market-based alternatives like Housing Choice Vouchers and the Low-Income Housing Tax Credit, while simultaneously dismantling the existing stock through the HOPE VI program, which has funded the demolition or removal of roughly 30% of the original federal public housing stock since the early 1990s (Schwartz 2021).

Yet the evidence for public housing's failure is mixed. A body of scholarship contends that the program has been unfairly maligned by its most notorious examples and argues that its poor reputation partly reflects broader social and economic challenges faced by American cities in the postwar period (Bauman 1994; Bloom, Umbach, and Vale 2015; Goetz 2013). Indeed, despite widespread criticism of the program, approximately two-thirds of public housing residents reported being "satisfied" or "very satisfied" with their housing as late as 1999 (Schwartz 2021). Some causal evidence from economists finds that living in public housing has either neutral or positive average effects on individual outcomes (Chaudhry and Eng 2024; Currie and Yelowitz 2000; Jacob 2004; Pollakowski et al. 2022), suggesting that the program's most notorious failures may not be representative.

At the same time, disentangling the local effects of public housing from pre-existing

neighborhood conditions remains challenging. Given that projects were often sited in already-distressed areas as part of slum clearance efforts, did public housing cause neighborhood decline, or simply reflect the targeting of struggling areas? These debates have practical stakes as cities grapple with housing affordability and revisit the question of what role public housing should play.

To shed light on these debates, I construct a new dataset containing the locations, construction dates, and characteristics for the near-universe of public housing projects built from the program's inception until 1973 by combining existing digitized records, administrative data, and newly collected archival sources. I combine these data with a newly constructed long-run panel of neighborhood characteristics from 1930 to 2010.

I first examine which neighborhood characteristics predicted public housing site selection. Did public housing specifically target Black neighborhoods, as some historical accounts suggest (e.g., Hirsch 1998; Rothstein 2017), or was it more broadly targeted toward poor and working-class neighborhoods? This question is central to both understanding the political economy of public housing and credibly estimating the effects of public housing construction on neighborhoods. I estimate linear probability models to explore how pre-existing neighborhood characteristics, including demographic, socioeconomic, housing, and institutional factors, predict the placement of public housing projects. Consistent with the program's slum clearance mandate, projects were more likely to be built in neighborhoods that were initially poorer and more populous, with lower homeownership rates and higher shares of children. Even conditional on these socioeconomic and housing characteristics, however, public housing was much more likely to be sited in neighborhoods with higher Black population shares, reflecting the racialized politics of mid-century housing policy. I also find that public housing was more likely to be built in neighborhoods that were later targeted by the urban renewal program, another major mid-century urban policy that disproportionately affected Black neighborhoods (LaVoice 2024).

I then estimate the short- and long-run effects of public housing construction on both recipient and surrounding neighborhoods. Following the approach used by Chetty, Diamond, et al. (2026) to study the effects of public housing demolitions, I use stacked matched difference-in-differences designs that compare each recipient neighborhood to neighborhoods that looked similar before public housing was built. First, I use nearest-neighbor propensity score matching to select three potential control neighborhoods for each recipient neighborhood based on initial neighborhood characteristics. Second, within each treatment cohort, I pool the selected controls and reweight them to more closely resemble the recipient neighborhoods. Finally, I stack the cohort panels and estimate event studies, comparing changes in recipient neighborhoods with changes in their weighted

controls. Identification stems from the fact that federal funding for public housing was limited, and local housing authorities could not build projects in every eligible neighborhood, creating variation in project placement across otherwise similar areas. This stacked framework also avoids econometric pitfalls in staggered adoption settings when treatment effects vary across cohorts and over time (Wing, Freedman, and Hollingsworth 2024). I apply an analogous design to tracts within one kilometer of recipient areas to estimate spatial spillovers into surrounding neighborhoods.

In the public housing ("recipient") neighborhoods, I find sizable increases in Black population shares, driven mainly by growth in the Black population: three decades after construction, the Black population share is 9.4 percentage points higher than in matched control neighborhoods. I also find sizable decreases in median rents (-16.2%) and incomes (-15.6%), along with declines in homeownership rates. These results, together with the site-selection results, suggest that while public housing targeted poorer neighborhoods with a higher share of Black residents, the projects' construction further accelerated racial transition and socioeconomic decline in these neighborhoods over the long run.

In nearby neighborhoods, I also find evidence of racial transition, with little evidence of long-run effects on incomes or rents. Using directly matched controls, the Black population share rises by 6.0 percentage points three decades after construction. Here, the shift is driven by declines in the white population, rather than increases in the Black population.

I then examine how these effects vary across project cohorts and characteristics to understand what drives the neighborhood changes. In recipient neighborhoods, the long-run effects of public housing are concentrated around early-generation family developments opened between 1940 and 1960, whereas elderly projects opened between 1960 and 1973 raised recipient Black shares by less, left median incomes unchanged, and caused no significant white flight from nearby tracts. Project size does not account for the larger effects of early family projects. Splitting early family developments at their median size yields nearly identical Black-share increases, and the differences in nearby white outflows and recipient incomes are not statistically significant. Instead, the effects closely track the racial makeup of a project's tenants. In recipient neighborhoods, the Black population share rose and median income fell most where a project's tenants transitioned to majority Black after opening, while white residents left both recipient and nearby neighborhoods only around projects whose tenants were or became majority Black.

Finally, I explore the potential implications of these neighborhood effects for long-run economic opportunity. To do so, I link my dataset to Opportunity Atlas data on the upward mobility and later incarceration rates of children from low-income families born between 1978 and 1983 (Chetty, Friedman, et al. 2018). I find that recipient tracts have lower

upward mobility and higher incarceration rates than their weighted controls. Conditioning on 1980 neighborhood characteristics removes most of the income-rank gap and more than half of the incarceration gap, suggesting that public housing-driven changes in neighborhood composition account for much of these differences. Nearby neighborhoods show the same pattern at about half the size, and their gaps also close once 1980 characteristics are held fixed.

This paper relates to several strands of the literature in urban and public economics and economic history. First, it contributes to work on the local effects of affordable housing programs (Baum-Snow and Marion 2009; Diamond and McQuade 2019) and neighborhood revitalization policies (Collins and Shester 2013; LaVoice 2024; Rossi-Hansberg, Sarte, and Owens 2010; Blanco and Neri 2025) by estimating the neighborhood effects of one of the largest urban policies in American history. More directly, studies of HOPE VI-era public housing demolitions find that removing distressed projects reduced nearby crime, increased housing demand, changed neighborhood racial and income composition, and generated uneven welfare effects across groups (Tach and Emory 2017; Blanco 2023; Aliprantis and Hartley 2015; Sandler 2017; Almagro, Chyn, and Stuart 2023). Related work following displaced children finds long-run gains in employment and earnings, especially for children moved out of the most distressed projects (Haltiwanger et al. 2024; Chyn 2018), and also finds that HOPE VI may have created higher opportunity neighborhoods (Chetty, Diamond, et al. 2026). This literature has focused on demolitions of the most distressed public housing projects, much of it in a single city (Chicago), and its estimates may not generalize to public housing construction. HOPE VI targeted the nation's most severely distressed projects, which may not represent the broader public housing stock. The effects of construction also likely depended on existing neighborhood conditions in ways that demolition effects did not, since new projects may have served as local amenities in some neighborhoods and disamenities in others. And effects observed in Chicago may not generalize to other cities.

My findings also complement earlier work in other social sciences on public housing and the concentration of poverty in central cities, which generally used data compiled from one or a few cities (e.g., Carter, Schill, and Wachter 1998; Massey and Kanaiaupuni 1993). Relative to this older literature, I use a dataset covering a much broader set of cities, projects, and time periods, and apply modern econometric techniques to estimate the causal effects of public housing construction on neighborhoods.

Most directly, this paper contributes to the literature on the effects of public housing construction. Shester (2013) uses digitized HUD data covering all federally funded projects built through 1973, with locations available only at the locality level (Section 4),

and finds that cities in the same state with more public housing construction saw declines in median property values, median family income, and population density. Shester, Allen, and Handy (2019) use the same data to study the rise in single motherhood across metropolitan areas. I build on this work by adding precise project locations to these data, which lets me study the neighborhood effects of construction.

My paper fits into an emerging literature on the neighborhood-level effects of public housing construction. Günnewig-Mönert (2025) studies the effects of public housing construction in New York City from 1930 to 2010 on neighborhood racial composition, rents, and welfare, and how these effects differed based on the design of the projects. My paper builds on this work by assembling project-level data for the entire country to characterize the public housing program more broadly. This scope matters because the New York City Housing Authority may have been an outlier among American housing authorities, having largely avoided the management failures and high-rise demolitions that defined the program elsewhere (Bloom 2008; Schwartz 2021). Estimates from New York City alone, therefore, may not be representative of the program as a whole. Two other recent papers study the initial effects of public housing construction during the program's first decades. Harris (2025) studies the local effects of public housing constructed from 1940 to 1960 and, using a shift-share IV approach, finds that public housing initially reduced the concentration of poverty in recipient neighborhoods. Blanco and Perdoni (2026) study the initial public housing constructed during the New Deal and find limited effects on broader neighborhood composition. By contrast, I examine the near-universe of mid-century projects and trace their long-run effects as the program evolved over time.

Finally, my paper contributes to the economic literature on the emergence and consequences of segregation in the United States. While prior research has documented migration responses to neighborhood demographic change (Shertzer and Walsh 2019; Boustan 2010), my paper addresses one way in which federal housing policy directly influenced neighborhood sorting. More broadly, this paper contributes to long-standing debates around the role that public policy has played in shaping racial and economic segregation in American cities (Rothstein 2017; Trounstine 2018; Boustan 2013; Logan and Parman 2025).

The paper proceeds as follows. Section 2 describes the history of the public housing program. Section 3 outlines the potential channels through which public housing may affect neighborhoods. Section 4 describes data sources and construction. Section 5 analyzes the determinants of public housing site selection. Section 6 presents the empirical strategy and estimates of neighborhood effects. Section 7 explores heterogeneity in these effects and examines potential mechanisms. Section 8 links public housing to long-run

economic opportunity using data from the Opportunity Atlas. Section 9 concludes.

2 Historical Background: The U.S. Public Housing Program

Public housing in the United States began during the Great Depression as part of New Deal efforts to reduce urban housing shortages and stimulate the economy. The Public Works Administration built the first projects, producing about 20,000 units before court challenges over land acquisition limited its scope (Jackson 1985). The program was greatly expanded and decentralized with the passage of the Housing Act of 1937 (the "Wagner-Steagall Act"), which promoted the creation of local Public Housing Authorities (PHAs) to build and oversee public housing with federal funds. Specifically, PHAs were responsible for choosing public housing locations, building the projects, and managing them. Under this federal-local framework, Washington financed capital construction costs by servicing local housing authority bonds, while PHAs were required to cover ongoing operating and maintenance expenses almost entirely through tenant rents (McCarty 2014). The 1937 Act's primary mandate was slum clearance and redevelopment rather than net housing supply expansion, reflecting contemporary beliefs that "slum" neighborhoods reduced nearby property values and caused poor health and behavior among residents (Meyerson and Banfield 1955). Accordingly, the law's "equivalent elimination" provision required localities to demolish or rehabilitate one unsafe or unsanitary dwelling for every newly constructed public housing unit (McCarty 2014).

The program expanded dramatically as part of a broad urban policy agenda following World War II. Severe postwar housing shortages—which were particularly acute for Black households—alongside accelerating suburbanization, white flight, and perceived urban "blight" forged a broad political coalition for urban redevelopment (von Hoffman 2000). This push culminated in the Housing Act of 1949. Title III of the Act authorized federal funding for an additional 810,000 units of public housing and significantly broadened the program's role. Beyond clearing slums and providing low-income housing, public housing became a primary relocation mechanism for households displaced by the Title I urban renewal program, who were granted statutory preference for public housing units (McCarty 2014).

The ambitious goals of the 1949 Act, however, quickly encountered political, demographic, and social obstacles during the 1950s and 1960s that transformed public housing into a deeply controversial policy. First, as described in Sections 2.1 and 5, intense local conflicts over race and site selection restricted where projects were ultimately built. Second, the socioeconomic and demographic profile of public housing tenants changed

significantly over time. Whereas early projects primarily housed working families and the "deserving poor," rapid suburban homeownership growth enabled white working-class households to exit public housing in the 1950s and 1960s, dramatically changing the projects' racial composition. Simultaneously, federal legislation tightened income eligibility limits and mandated that PHAs prioritize the neediest applicants, concentrating poverty within the projects (Schwartz 2021).

Third, physical design flaws and a worsening fiscal structure undermined project maintenance. To avoid competing with private housing and comply with strict statutory per-unit cost limits, projects were intentionally austere and inexpensive, making them prone to rapid deterioration (Schwartz 2021). As middle-income tenants exited and tenant incomes fell, many public housing authorities struggled to fund basic upkeep. Furthermore, the 1969 Brooke Amendment capped tenant rents at 25% of income, further limiting public housing authorities' ability to fund necessary maintenance with revenues from tenant rents (Hunt 2018). Figure 1 illustrates this growing fiscal squeeze using newly digitized national operating statements from the Housing and Home Finance Agency (HHFA) and HUD from 1941 to 1977, showing that monthly rental receipts comfortably exceeded operating costs in the 1940s and 1950s before falling below operating expenses following the 1969 Brooke Amendment.

Mounting public backlash against public housing reshaped federal housing policy after the 1960s. A 1971 report by the Nixon Administration stated that "drab, monolithic housing projects, largely segregated…still stand in our cities as prisons of the poor" (Orlebeke 2000), and in 1973, President Nixon declared a moratorium on subsidies for traditional public housing. The Housing and Community Development Act of 1974 later established the Section 8 program to subsidize low-income tenants in privately owned rental housing, and in 1986 the Low-Income Housing Tax Credit (LIHTC) program followed. By the early 1990s, deferred maintenance, extreme concentration of poverty, and elevated crime in the most distressed public housing projects prompted Congress to launch the HOPE VI program in 1992, which provided federal grants to demolish or redevelop severely distressed projects and transform them into mixed-income developments. Meanwhile, the Faircloth Amendment of 1998 capped the total number of units that any housing authority could operate at its 1999 levels, effectively ending the expansion of federal public housing.

2.1 Race, Segregation, and Public Housing

Race and segregation emerged as central and contentious issues throughout the program's history. From the outset, the Public Works Administration followed a so-called "neighborhood composition rule," formally segregating projects based on the demographics of the neighborhoods where they were built. While framed as a neutral policy intended to maintain "neighborhood stability," in practice, the rule institutionalized segregation and, in some cases, imposed it on previously integrated communities. Prominent accounts, such as Rothstein (2017), argue that federal officials and local authorities used public housing to dismantle integrated neighborhoods and replace them with segregated projects, as in Atlanta's all-white Techwood Homes in 1936.[1] Many large public housing authorities continued to follow an explicit racial assignment rule until the 1950s and 1960s.[2] However, even after these legal changes, projects in many cities remained highly segregated in practice (Bickford and Massey 1991). Even as legal mandates ended formal segregation, the tenant populations inside public housing underwent a dramatic racial transition. Figure 2 documents the changing racial composition of public housing across major urban housing authorities using new data described in Section 4.1. Between 1958 and 1977, the Black share of public housing tenants rose sharply in nearly every major city.

Racial politics also profoundly shaped project site selection. Contemporaneous debates make clear that policymakers fully understood the program's potential to either entrench or reduce existing residential segregation patterns (Hirsch 2000). In Chicago in the 1950s, the Chicago Housing Authority had initially planned to build projects throughout the city, but strong opposition from white neighborhoods led to the concentration of projects in predominantly Black areas on the South and West Sides (Meyerson and Banfield 1955; Hirsch 1998). Similar conflicts occurred in other major cities, with white residents and politicians successfully blocking projects in their neighborhoods. I examine a concrete example of this dynamic in Philadelphia in Section 5.2, comparing proposed-but-not-selected sites to built project locations, and test in Section 5.1 whether these racial site selection dynamics held systematically nationwide.

[1] By contrast, recent empirical work by Blanco and Perdoni (2026) finds limited evidence that New Deal-era public housing projects systematically altered broader neighborhood racial composition on average.

[2] Legal challenges against segregation in public housing began in the mid-1950s, including the U.S. Supreme Court's 1954 decision to allow California courts to ban segregation in San Francisco's housing projects. However, federal law did not formally prohibit race-based segregation in public housing until the 1960s. An executive order in 1962 directed federal agencies to prevent racial discrimination in federally owned housing, and Title VI of the Civil Rights Act of 1964 prohibited discrimination in any program receiving federal financial assistance (McCarty 2014).

2.2 Motivating Example: Hunters Point in San Francisco

The history of Bayview–Hunters Point in San Francisco provides a clear example of how public housing construction can change a neighborhood's long-run trajectory. Before World War II, the area was a racially mixed neighborhood on the city's periphery. During the war, the federal government built thousands of temporary housing units there to accommodate defense workers, which led to an initial influx of African American families into the neighborhood. After the war, rather than dismantling these temporary units, local and federal officials converted many of them into permanent public housing. Throughout the 1950s and 1960s, the San Francisco Housing Authority replaced and expanded these units into large permanent public housing developments. This expansion coincided with a rapid neighborhood racial turnover: between 1940 and 1970, white households exited the area and surrounding tracts, and the community became heavily Black and racially segregated (Figure 3). The rest of this paper tests whether the patterns seen in Hunters Point held systematically across the United States.

3 Potential Neighborhood Effects of Public Housing

To fix ideas, I briefly outline the main channels through which the construction of public housing projects might have affected neighborhood outcomes in the short and long run. First, public housing construction may have mechanically altered the racial and socioeconomic composition of neighborhoods, because tenant selection policies, income limits, and subsidized rents determined who occupied these projects and thus the demographic makeup of the areas where they were built. Second, these compositional changes may have triggered endogenous sorting by private households: if individuals have preferences for their neighbors' race or income, the arrival of public housing residents might have prompted some existing residents to relocate. Prominent historical accounts such as Jackson (1985) argue that public housing drove "white flight" from central cities, though empirical evidence on this point remains limited.

Third, the physical structure of public housing projects could create built-environment externalities: replacing substandard or vacant structures with new housing could improve neighborhood quality and desirability, as shown for private housing upgrades by Rossi-Hansberg, Sarte, and Owens (2010) and for subsidized housing by Ellen et al. (2007). Conversely, large superblocks or architecturally incongruous high-rise towers might be viewed as local disamenities. Indeed, criticism of the mid-century program has often focused on the design of the projects (Jacobs 1961; Newman 1972). Fourth, public housing

construction may generate broader social and market externalities. A frequent concern is that concentrated poverty within large projects generates social disorder and elevated crime, with potential spillovers into surrounding neighborhoods (Sandler 2017). The direction of these effects may vary across contexts: modern evidence on the LIHTC program suggests that subsidized housing investments can be a positive amenity in distressed areas but a disamenity in more affluent ones (Diamond and McQuade 2019).

4 Data

4.1 Public Housing Data

A significant challenge in studying the history of public housing is the lack of a comprehensive public dataset that includes project lists, construction dates, and precise project locations. Consequently, previous research on the neighborhood effects of public housing has largely been limited in scope or restricted to a small number of cities where researchers obtained this information directly from housing authorities. These data limitations have hindered broader historical analyses of the public housing program, as previous work has acknowledged (Ellen et al. 2007; Hunt 2018). The administrative data that do exist were largely compiled in recent decades and therefore often omit projects demolished earlier or otherwise ceased to exist—disproportionately among the most distressed developments. I address this gap by assembling what I believe is the most complete dataset of mid-century public housing by combining information from a series of administrative datasets with previously digitized and newly digitized sources.

The first source I use is the *Consolidated Development Directory* (CDD), published by the U.S. Department of Housing and Urban Development (HUD) in 1973 and digitized by Shester (2013). The CDD lists all federally funded public housing projects that existed in 1973 and reports project characteristics such as unit counts and project completion dates.[3] The CDD lacks detailed project locations, so previous work using these data has focused on aggregate city- and county-level outcomes (Shester 2013; Shester, Allen, and Handy 2019). These data also include federal project codes, which I use to link these records to later HUD datasets with building addresses and project coordinates.

I supplement the CDD with historical project inventories obtained from libraries, hous-

[3]The directory also records leased housing projects, privately owned housing that local authorities leased and subleased to low-income families under the Section 23 program, which began in 1965. I exclude these leased projects from my analysis. A description of the Section 23 program can be found in National Housing Law Project (2012). I also exclude housing built under the federal Indian Housing Program, which tribal housing authorities administered separately.

ing authorities, and public records requests. I collected these sources for two reasons. First, the CDD covers only the federal program: little data exist on city- and state-funded public housing, and New York City, in particular, has had a notable number of city- and state-funded projects. Second, in cities where both sources exist, the inventories provide an independent check on the CDD's project lists and completion dates.

I locate projects by compiling a national dataset of individual public housing building addresses from several sources: the building-level address file underlying HUD's *Subsidized Housing Projects' Geographic Codes* (Form HUD-951), which records the addresses of roughly 288,000 public housing buildings surveyed in 1988–89 (U.S. Department of Housing and Urban Development 1996); HUD's National Geospatial Data Asset (NGDA) building inventory in 2023, from which roughly 106,000 currently existing buildings enter my address dataset[4]; the 2016 New York City Housing Authority *Address Guide*; and a list of the addresses of all public housing projects that were owned by the Chicago Housing Authority, provided to me by D. Bradford Hunt. I also digitized the building addresses for the Pruitt-Igoe project in St. Louis from its 1964 site plan. I geocode these addresses using ESRI's StreetMap Premium geocoder and link buildings to the CDD projects using the federal project codes described above, along with name-based matching in New York City. Building-level addresses let me capture the full project footprint rather than just a single project address, which is important because some mid-century projects were quite large. For projects lacking a usable building address, I use project-level coordinates from HUD's Picture of Subsidized Households (PSH) datasets from 1997 and 2000, which report coordinates for federally funded housing projects from 1997 onward.

Overall, I identify the locations of almost the entire universe of mid-century public housing in the United States. I geolocate 99% of the total completed public housing units in the CDD, and the projects that I am unable to locate are predominantly small, with a median of 24 units. In my final analysis sample, 82% of projects are located through exact building addresses, and the remainder through project-level coordinates.

I supplement the project inventory with information on each project's tenants: how many residents it housed, their racial composition over time, and whether the project was designated for families or elderly. I ultimately use the resident counts to estimate each neighborhood's private population in each decade by subtracting the estimated public housing population from its census population (Section 6), and the tenant composition of the projects to explore heterogeneity in the effects of public housing (Section 7).

[4]HUD revamped its project codes in 2008, meaning that a direct match between NGDA project codes and CDD project codes may mismatch projects. I construct a crosswalk between the 2007 and 2008 project codes to do this mapping, described in Appendix A.

I measure each project's population using HUD's 1977 Picture of Subsidized Households (PSH), which reports tenant households by race for most federally funded projects as of 1971–77 (Burke 1997).[5] Because the HUD data cover only federally funded projects, I use the New York City Housing Authority's 1973 data book for the city's city- and state-funded projects, and the Chicago Housing Authority's 1973 annual report for Chicago projects missing from the 1977 PSH.

I also compile new data on how the racial composition of public housing changed over time from three kinds of sources (Appendix A.3). The first is published tenant counts by race, from housing authority reports, tenant surveys, civil rights hearings, and court records. These include the Public Housing Administration's *Trends Toward Open Occupancy*, whose five editions between 1955 and 1963 report Black-occupied units for 2,656 project listings, 2,422 of which I link to projects in my inventory. For New York City, I use the housing authority's counts of families by race at initial occupancy and in December 1971, digitized and shared with me by Max Günnewig-Mönert (Günnewig-Mönert 2025). The second source is the 1940 and 1950 full-count censuses, which I match to project building addresses. The third is the official designation of a project as white or Black while its city formally segregated public housing, which I take from housing authority lists, congressional hearings, court records, and historic preservation filings.

Finally, I also identify whether the project was designated for families, only elderly tenants, or both. I take this designation from the 1977 Picture of Subsidized Households when available, then from the elderly unit counts in HUD's 1973 directory, and otherwise from housing authorities' own project lists and a few state and federal reports (Appendix A.4).

As a retrospective measure of project distress, I also link projects to HUD's roster of HOPE VI revitalization grants awarded between 1993 and 2010, obtained from the site list published by Chetty, Diamond, et al. (2026).[6] I match award development names to my projects within each state and city and resolve ambiguous cases through manual review. Eventual HOPE VI selection serves as a retrospective marker of projects that later became especially distressed and received federal funding for demolition or redevelopment.

[5]These data report the number of subsidized households by race in each project, but do not include direct population counts. In Appendix A, I describe how I convert these household counts to population counts. These data were originally shared with me by Yana Kucheva (Kucheva 2013).

[6]Available at https://opportunityinsights.org/paper/hopevi/

4.2 Neighborhood Data

To study the neighborhood effects of public housing construction, I construct a panel dataset of neighborhood characteristics from 1930 to 2010.

First, I collect a variety of outcomes from the 1930 to 2010 decennial census tract tables, which I download from IPUMS NHGIS (Manson et al. 2022). These data include tract-level population counts by race (white, Black, and other), age structure (share under 18 and share 65 and older), several socioeconomic measures (median income, high school graduation rates, labor force participation, and unemployment rates), and housing market measures (median contract rents, homeownership rates, and the share of housing units needing major repairs).[7] I also measure each tract's distance to the nearest central business district (CBD), using the CBD indicator published in the 1980 census tract tables.

I harmonize all observations to 1950 census tract boundaries using an area-reweighting procedure (Eckert et al. 2020). Anchoring the panel to an early baseline geography prevents endogenous, public-housing-induced tract boundary changes from confounding the estimates (Glaeser, Gyourko, and Neiszner 2025). I use 1950 rather than 1940 boundaries because they cover more locations and to avoid issues with the 1940 New York City NHGIS tract shapefiles, which represent larger Health Districts rather than Census tracts. Appendix B.2 describes the tract harmonization process in more detail. I group census tracts into core-based statistical areas (CBSAs) using the Office of Management and Budget's June 2003 delineations, based on Census 2000 population and commuting data (Office of Management and Budget 2003).

I supplement the published tract-level data with full-count Census microdata from 1930 and 1940 (Ruggles et al. 2022). This lets me expand geographic coverage beyond cities and neighborhoods with published tract-level data in 1930 and 1940 and construct neighborhood income measures not available in those years' census tract tables. The full-count microdata identify enumeration districts, which are smaller than census tracts, so I aggregate individuals to enumeration districts and map these to 1950 tracts in two ways. Where the Urban Transitions Project provides 1930 and 1940 enumeration district shapefiles (Logan, Minca, et al. 2024), I allocate district totals to 1950 tracts by area. For 1940 districts outside those shapefiles but in cities that had census tracts in 1940, I use newly scraped National Archives descriptions, which record the census tract containing each district, to aggregate the microdata to 1940 tracts and then concord them to 1950 tracts. The 1930 Census did not collect income data, so for that year, I sum the machine-learning-adjusted

[7] I draw the share of units in need of major repairs from the 1940 tract tables. I use this measure only for 1940, as it is not consistently available in other decades.

occupation scores developed by Saavedra and Twinam (2020) across family members. For 1940, I construct family wage income by summing individual wage and salary earnings. In both years, I count families by income bracket within each enumeration district, allocate these counts to 1950 tracts by area, and compute each tract's median from the resulting distributions, as I do for the published tables (Appendix B.2.1). I express all monetary values in 2000-dollar terms using the U.S. CPI from Officer and Williamson (2025).

I supplement the Census panel with three sets of historical geographic covariates. First, I incorporate the digitized data and shapefiles of the Home Owners' Loan Corporation (HOLC) "redlining" maps drawn in the late 1930s, made available by *Mapping Inequality* (Nelson and Winling 2023). These maps graded neighborhoods from "A" (best) to "D" (hazardous) and capture perceived credit risk and potential discrimination.[8] Following Weiwu (2025), I classify a neighborhood as "redlined" if at least 80% of the area is designated as "hazardous" (HOLC grade D). Second, I incorporate data on the locations of U.S. urban renewal projects (1955–1966) from *Renewing Inequality* (Nelson and Ayers 2025). Urban renewal, established under Title I of the 1949 Housing Act, provided federal subsidies for cities to acquire and clear "blighted" neighborhoods. Urban renewal was closely intertwined with public housing: clearance projects often created sites for new developments or displaced families that public housing was intended to rehouse (von Hoffman 2000).[9] To account for potentially confounding effects of urban renewal and to explore the relationship between the two programs, I overlay these maps onto 1950 census tract boundaries and classify a tract as an "urban renewal" tract if at least 5% of its area overlaps with an urban renewal project. Finally, I incorporate data on the locations of interstate highways as of 1996 from Weiwu (2025). Interstate highway construction began in the late 1950s and has been accused of disproportionately displacing minority communities (Rose and Mohl 2012). I calculate the distance from each tract to the nearest interstate to examine the relationship between highway locations and public housing placement, and to assess the potential confounding effects of highway construction on neighborhood change.

Finally, I measure long-run individual outcomes using the Opportunity Atlas (Chetty, Friedman, et al. 2018), which reports adult outcomes by childhood census tract for children born between 1978 and 1983. I use two outcomes, pooling across race and gender:

[8] Although many scholars have argued that these maps institutionalized redlining and curtailed investment in minority neighborhoods (Aaronson, Hartley, and Mazumder 2021), recent evidence complicates this view. Fishback et al. (2022) show that the maps largely reflected existing patterns of racial and socioeconomic segregation rather than shaping new lending decisions.

[9] Recent research finds that Black neighborhoods were two to three times more likely to receive urban renewal projects than white neighborhoods, and that urban renewal areas experienced declines in housing and population density alongside rising rents and incomes (LaVoice 2024).

mean adult household income rank, measured in 2014–2015, and the incarceration rate on April 1, 2010. For both outcomes, I use the Atlas's estimates for children whose parental income was at the 25th percentile of the national income distribution. I refer to these children as "low-income children." These outcomes are available at the 2010 Census tract level, so I harmonize them to 1950 census tracts using my geographic crosswalks and the Opportunity Atlas's counts of children in each group.

4.3 Defining the treatment

To study the effects of public housing construction on neighborhoods, I define a set of recipient neighborhoods that received public housing projects, as well as a set of nearby neighborhoods that did not receive public housing directly but were most likely to experience geographic spillovers due to their proximity to a public housing project.

First, to ensure that a project represents a meaningful potential intervention at the census tract level, I exclude projects with fewer than 50 housing units. I also exclude "scattered site" projects, which are often listed under a single project identifier but actually represent housing units located throughout a city. I identify scattered sites by name, or, for projects with at least five geocoded buildings, when the median distance of the buildings from their own median location exceeds one kilometer.[10]

I define recipient tracts based on the locations of the individual buildings that make up each project. When a project's buildings are geocoded, I assign each building to the tract it falls within, and consider a tract a recipient tract if it contains at least one building from a qualifying public housing project. For projects where I was only able to geocode a single project-level coordinate, I consider any tract intersecting a 100-meter buffer around it as a recipient tract. This buffer captures the fact that some projects are large enough to straddle multiple tracts. Under either rule, a project may span several tracts. When it does, I allocate its units and population across them in proportion to the number of buildings in each tract. For projects assigned based on the project-level coordinate, I divide them evenly.

Since the Census data are decennial, I assign treatment timing by decade. Following the literature, I use the project's completion date as the treatment date (Asquith, Mast, and Reed 2023). For CDD projects, I use the year of HUD's "date of full availability," the month in which substantially all of a project's units became available for occupancy. For projects from the city inventories, I use the completion year reported in the inventory. I assign a

[10]This threshold was chosen by hand, and does not drop ultra-large contiguous developments in the data, such as the Robert Taylor Homes corridor in Chicago.

recipient tract to cohort t if its first qualifying project was completed between April of year $t-10$ and March of year t, since the censuses are enumerated on April 1 of each year. This timing convention ensures that the treatment occurs after the pre-treatment observation but before the post-treatment observation in the panel. For tracts that receive multiple public housing projects over time, treatment timing is determined by the first project that meets the size threshold. One other challenge with these data is that, for defense housing built during World War II and later conveyed to local housing authorities, HUD's data often records the date of conveyance rather than the opening date. For these projects, I instead use the occupancy dates printed in another source, the National Park Service's *Public Housing in the United States, 1933–1949*.

To examine spatial spillovers, I define a nearby tract as one that never received a qualifying public housing project but whose boundary lies within one kilometer of at least one building of a qualifying project. For projects located only by project-level coordinates, I measure the distance to those coordinates instead. Nearby tracts inherit the earliest treatment year among their recipient neighbors, and a tract within one kilometer of several projects is assigned the completion year of the earliest project. I adopt the one-kilometer threshold based on prior literature showing the geographic extent of spatial spillovers from public housing redevelopment (Blanco and Neri 2025). Figure 4 illustrates the recipient and nearby classification for Chicago.

4.4 Sample Construction

My empirical analysis focuses on projects built between April 1940 and 1973, with neighborhood outcomes measured from 1930 to 2010. I restrict the sample in several ways to exclude outlier tracts and to ensure every tract has enough pre-treatment data for the difference-in-differences analysis in Section 6.

Table 1 reports the sample restrictions and resulting attrition. I first drop neighborhoods that were initially sparsely populated — those with fewer than 250 residents in 1940 — and tracts with more than 8 people per household in 1940, which are tracts that seem mostly dominated by institutions or group quarters like hospitals and rooming houses. I also drop geographic outliers larger than 10 km^2, which are typically peripheral suburban tracts. I then exclude tracts whose first qualifying project was completed by the April 1940 census, so that every recipient tract has two pre-treatment observations. Finally, while I do not require a balanced panel over the entire period to avoid unnecessarily losing data, I drop tracts that are missing data over the relevant analysis period. For recipient and nearby tracts, I require nonmissing values of the main variables in two pre-treatment

decades and in every decade after it through 2010. For never-treated tracts, I require complete data for two pre-treatment decades for at least one treatment cohort through 2010. In practice, this means keeping never-treated tracts with complete data from 1960 through 2010.

The resulting sample includes 9,759 census tracts across 59 CBSAs, representing 35.3 million people in 1940 (approximately 27% of the U.S. population). This sample contains 1,213 public housing projects with 438,003 total units located in 1,186 recipient tracts. Figure 5 shows the geographic distribution of CBSAs included in the analysis. These sample restrictions have implications for the external validity of my analysis. First, because of data limitations, my analysis focuses on neighborhoods in major metropolitan areas and excludes public housing built in rural areas or smaller cities. Second, by excluding projects with fewer than 50 units, my results also do not speak to the effects of smaller-scale or scattered-site public housing. Finally, my results exclude the earliest public housing projects, including the New Deal projects studied by Blanco and Perdoni (2026). Appendix Table A1 compares included and excluded projects and shows that the analysis sample disproportionately includes larger developments and projects in the Northeast and Midwest.

5 Site Selection

In this section, I study the placement of the public housing projects in my sample. These dynamics matter for several reasons. First, the site-selection process for mid-century public housing was a significant criticism of the program. Historical case studies in several cities suggest that projects were often targeted to poorer and minority neighborhoods, both because of the program's slum clearance goals and because of local backlash in white neighborhoods against construction of projects (Meyerson and Banfield 1955; Bauman 1994; Sugrue 2005). Second, site selection dynamics have had major legal consequences: civil rights lawsuits in several cities have alleged that public housing site selection was discriminatory and in violation of civil rights law, most famously *Gautreaux v. Chicago Housing Authority* (1966), but also in cities like Dallas (*Walker v. HUD*, 1985) and Baltimore (*Thompson v. HUD*, 2005). Still, there has been little systematic evidence on the nationwide patterns of public housing site selection. Finally, understanding these site-selection dynamics is important for estimating the neighborhood effects of public housing and interpreting those effects.

To begin, in Section 5.1, I estimate which pre-existing neighborhood characteristics predict the eventual locations of public housing projects. I also test whether the placement

of public housing projects was related to the locations of other mid-century urban policies, particularly urban renewal and the interstate highway system. Then, in Section 5.2, I zoom in on a particular case study of site selection by examining the locations of proposed-but-not-built public housing in Philadelphia. This example illustrates the political dynamics around site selection decisions.

5.1 Where were the projects built?

Table 2 shows the raw comparison of 1940 baseline characteristics between census tracts that eventually received a qualifying public housing project and those that did not. Recipient neighborhoods had substantially higher Black population shares, lower median incomes and rents, higher unemployment rates, a higher share of homes needing major repairs, and were located closer to a central business district (CBD).

To formally test which neighborhood characteristics predict public housing placement, I estimate linear probability models relating pre-treatment tract characteristics to the likelihood of ever receiving a public housing project. Specifically, I estimate:

$$\text{Recipient}_{im} = \gamma_m + \boldsymbol{X}'_{i,1940}\boldsymbol{\beta} + \boldsymbol{Z}'_i\boldsymbol{\delta} + \epsilon_{im} \tag{1}$$

where Recipient_{im} is an indicator for whether tract i in CBSA m received a qualifying public housing project between April 1940 and 1973, using the project eligibility and tract assignment rules described in Section 4.3. The vector $\boldsymbol{X}_{i,1940}$ contains 1940 neighborhood characteristics, and $\boldsymbol{Z}_i$ includes distance to the nearest interstate highway and an indicator for whether a tract was treated by urban renewal. CBSA fixed effects γ_m absorb differences in average placement rates across metropolitan areas, so the estimates compare neighborhoods within the same CBSA.

Columns (1)–(3) of Table 3 estimate the model without the $\boldsymbol{Z}_i$ controls, while Column (4) adds them to examine whether the relationship between 1940 characteristics and public housing placement is robust to controlling for subsequent federal interventions. I adjust standard errors for spatial correlation following Conley (1999), allowing for correlation of residuals across census tracts within a 2-kilometer radius.

I select covariates based on historical narratives of public housing site selection and the program's stated intentions. First, as described in Section 2, the public housing program was largely intended as a "slum clearance" program, so we might expect that neighborhoods with lower socioeconomic status and worse housing market conditions would be more likely to receive public housing. Second, historical case studies in multiple cities have documented the key role of race in determining where public housing was built (e.g.,

Hirsch 1998; Bauman 1994), so I include the Black population share as a key predictor of public housing placement. I also include the HOLC redlining designation to capture long-term patterns of racial segregation, discrimination, and disinvestment.

The results from these linear probability models are shown in Columns (1)–(4) in Table 3. Column (1) shows a parsimonious model with several key neighborhood characteristics, while Column (2) shows a more saturated model including all neighborhood characteristics, including homeownership rates and age composition. Column (3) adds the share of housing units deemed in need of major repairs in 1940, which is missing for about 5% of tracts. Consistent with the historical narrative, I find that public housing projects tended to be targeted toward poorer, minority neighborhoods: tracts that were initially more populous and had higher Black population shares, lower median incomes, and lower labor force participation rates were significantly more likely to receive public housing, as were tracts with higher unemployment rates at the 10% level. Neighborhoods with lower homeownership rates and a higher share of children were also more likely to receive public housing. The coefficients on redlining and median rent are not statistically distinguishable from zero in any column, and the housing disrepair coefficient in Column (3) is significant only at the 10% level. These findings reflect the slum clearance motivation of the public housing program and are also consistent with historical accounts emphasizing the racial targeting of public housing site selection: even after controlling for local economic, housing market, and demographic conditions, the Black population share remains a strong and significant predictor of public housing placement.

Column (4) additionally tests whether neighborhoods that ultimately received public housing were more likely to be affected by two other transformative mid-century urban policies: urban renewal and the interstate highway system. Evidence and historical narrative suggest that these programs may have affected many of the neighborhoods that were also targeted for public housing. The urban renewal program, in particular, was directly related to public housing in many cities, as public housing projects were used to house individuals displaced by urban renewal (Bauman 1994; Hirsch 1998). Neighborhoods that received public housing were also much more likely to be affected by urban renewal, whereas proximity to an interstate highway was not significantly related to public housing placement. This result further confirms the interplay between these two programs and motivates accounting for urban renewal in my empirical strategy in Section 6.4.[11]

Quantitatively, these effects are sizable. The baseline probability that a census tract in

[11]These results are consistent with similar exercises in Harris (2025) and Massey and Kanaiaupuni (1993).

the sample received a public housing project between April 1940 and 1973 is 13%. Based on the estimates in Column (4), a one standard deviation increase in Black population share is associated with a 4.6 percentage point (35.1%) higher public housing selection probability, a one standard deviation increase in unemployment rate with a 1.3 percentage point (9.8%) higher probability, and a one standard deviation decrease in log median income with a 3.7 percentage point (28.4%) higher probability. Finally, neighborhoods that were eventually targeted by urban renewal were 6.9 percentage points (53%) more likely to receive public housing.

5.2 A Philadelphia Site Selection Case Study

Historical accounts of public housing site selection have highlighted the political battles that surrounded the placement of projects in particular cities, for example, in Chicago (Meyerson and Banfield 1955), Philadelphia (Bauman 1994), and Detroit (Sugrue 2005). These accounts have emphasized the conflict between public housing authorities and white working-class neighborhoods that resisted the construction of public housing projects in their communities. Here, I present a case study of site selection in Philadelphia, drawing on the historical treatment and maps from Bauman (1987). This case study is illustrative of the political dynamics that shaped site selection decisions in many cities (Hunt 2018).

Following the 1954 Housing Act, the Philadelphia Housing Authority proposed 21 public housing projects across the city. Upon announcement, many of these proposed projects faced significant backlash from local white communities, leading to their eventual cancellation. I located these proposed sites by georeferencing in QGIS the historical map from Bauman (1987) shown in Figure 6, aligning it with a modern basemap using landmarks such as major roads and rivers, and matched the sites to 1950 census tracts. Fourteen of the 21 proposed sites were never built; they fall in 12 census tracts, which I classify as proposed-but-not-built neighborhoods.

Table 4 compares the baseline characteristics of these proposed-but-not-built tracts with those of actual public housing locations in Philadelphia, using all Philadelphia tracts with 1940 data as a citywide benchmark. The two groups differed greatly. Most notably, these proposed-but-not-built project locations had very low initial Black population shares. They were also further from the CBD, less populous, and had lower unemployment rates. These stark baseline differences illustrate how white neighborhoods successfully resisted public housing, and demonstrate why an empirical design using proposed-but-not-built projects as controls would not provide a credible counterfactual for built sites. Appendix Table A3 repeats the comparison for Chicago, using the sites the housing

authority proposed for family public housing between 1950 and 1966 and did not build, as listed in its 1968 affidavit in *Gautreaux v. Chicago Housing Authority*, and finds the same gap in Black population share.

6 The Effect of Public Housing Construction on Neighborhoods

In this section, I estimate the short- and long-run effects of public housing on both recipient and surrounding neighborhoods. I first describe the research design and then present the empirical estimates.

6.1 Research Design

6.1.1 Recipient Neighborhoods

The ideal experiment to study the effects of public housing construction on neighborhoods would be for public housing authorities to randomly select neighborhoods in each city to receive a public housing project. Then, one could simply compare the recipient and non-recipient neighborhoods over time. However, as shown in previous sections, public housing site selection was far from random. To approximate the ideal experiment, my primary empirical strategy is a series of difference-in-differences designs with matched controls, comparing recipient neighborhoods to neighborhoods that were initially observationally similar on the variables that predicted site selection. This design is similar in spirit to that used by Chetty, Diamond, et al. (2026) to study the effects of HOPE VI demolitions on neighborhoods.

Step 1: Defining the Donor Pool. First, I restrict the potential donor pool to census tracts that never received public housing and are never within one kilometer of a qualifying public housing project, as defined in Section 4. Excluding these surrounding neighborhoods will allow me to separately estimate the geographic spillovers of public housing while reducing concerns that geographic spillovers contaminate the control group.

I then restrict potential donors geographically. I use three main donor-geography rules, which trade off local comparability against observable neighborhood similarity. The same-division specification restricts selected donors to the recipient tract's Census division. The same-region specification permits donor tracts throughout the same Census region. Finally, the national specification permits donors from any metropolitan area. I report results using all three donor pools. Same-division controls are more likely to experience

similar local economic and policy shocks, while broader donor pools provide more potential controls and generally exhibit greater covariate overlap.

Step 2: Selecting comparable control neighborhoods. I next select neighborhoods that are similar to recipient neighborhoods in the preceding decades using nearest neighbor matching. For each treatment cohort and donor-geography rule, I estimate a logistic regression for treatment status based on pre-existing neighborhood characteristics. I include the main, statistically significant predictors of site selection from Section 5, as well as several other variables that were not statistically significant in the fully saturated site selection model, but are likely related to both public housing placement and subsequent neighborhood change. In particular, I match on log total population, Black population (in inverse hyperbolic sine), Black population share, log median income, log median rent, unemployment rate, labor force participation rate, homeownership rate, and the share of the population under 18, each measured in the two most recent pre-treatment decades. I also match on (time-invariant) redlining status and distance from the central business district. Using the estimated propensity scores from this regression, I select the three census tracts with the closest propensity scores as the controls for each recipient tract, with replacement.[12] A control may therefore be selected for several different tracts. Matching each cohort separately uses the pre-treatment characteristics relevant to that cohort. For example, for recipient tracts in the 1950 cohort, I estimate propensity scores based on neighborhood characteristics measured in 1940 and 1930.

Step 3: Pooling donors and constructing weights. Within each treatment cohort, I take the union of all distinct controls selected for at least one recipient tract, so that a control tract that was selected multiple times appears only once for that cohort. Recipient tracts are unique to a particular cohort, but control tracts can appear in multiple cohorts.

The matched control neighborhoods resulting from nearest-neighbor matching are more similar to the recipient neighborhoods in terms of observable characteristics, but they are not necessarily perfectly balanced on their pre-treatment characteristics. So, following Chetty, Diamond, et al. (2026), I re-estimate the same logistic regression in the matched sample, again separately for each treatment cohort, to adjust for the remaining differences between the recipient tracts and selected controls. Recipient tracts receive weight one, and control tracts i in treatment cohort g with fitted probability $\hat{p}_{ig}$ receive a preliminary weight equal to their estimated odds of recipient status:

$$\widetilde{w}_{ig} = \frac{\hat{p}_{ig}}{1 - \hat{p}_{ig}}$$

[12] Results are similar when using a different number of controls.

Thus, among the selected controls, greater weight is placed on neighborhoods whose pre-treatment characteristics more closely resemble those of the recipient neighborhoods.

Because my donor selection requires exact matching by donor geography, I rescale the preliminary control weights within each donor-geography cell. Let N_{cg}^{R} denote the number of recipient tracts in each donor-geography cell c and treatment cohort g, and let $\widetilde{W}_{cg}^{C}$ denote the sum of the preliminary weights assigned to controls in that cell. Each selected control tract then receives the final weight:

$$\frac{N_{cg}^{R}}{\widetilde{W}_{cg}^{C}}\widetilde{w}_{ig}.$$

The normalization factor $N_{cg}^{R}/\widetilde{W}_{cg}^{C}$ rescales all control weights within a cell so that their total weight equals the number of recipient tracts in that cell. This preserves the distribution of recipient tracts across donor-geography cells, while retaining the relative propensity-score weights of controls within each cell (see e.g., Iacus, King, and Porro 2012).

Table 5 compares recipient neighborhoods with the full national donor pool, as well as the matched, reweighted controls under each donor-geography rule. Standardized mean differences show the balance between each control group and the recipient tracts. Matching and reweighting substantially reduce differences in neighborhood characteristics, both for variables used in the matching and for those not included. One substantial remaining difference concerns later exposure to urban renewal, which is much more prevalent in recipient tracts compared to matched control areas. This overlap raises the possibility that some estimated neighborhood changes reflect the combined effect of the two programs. Section 6.4 compares results for neighborhoods with and without exposure to urban renewal.

Step 4: Stacked Difference-in-Differences Estimation. I then use a weighted stacked difference-in-differences estimator to compare changes in outcomes in the recipient neighborhoods with changes in the reweighted control neighborhoods before and after public housing construction. For each treatment cohort, I construct a panel containing the recipient tracts in that cohort and their selected controls, along with the weights constructed in Step 3, from 1930 to 2010. I assign the controls the treatment year of the corresponding cohort, and then stack the cohort-specific panels. A control tract may appear in multiple cohort stacks but receive a separate weight and comparison year in each stack.

I then estimate weighted event-study models using the matched samples and weights described above. For most of my outcome variables, I estimate the following linear specifications using weighted least squares:

$$y_{igt} = \alpha_{ig} + \lambda_{gt} + \sum_{\tau \neq -1} \beta_\tau (D^\tau_{igt} \times R_{ig}) + \varepsilon_{igt}, \tag{2}$$

where y_{igt} is the outcome in tract i assigned to treatment cohort g in decade t, R_{ig} indicates that tract i is a recipient tract in cohort g, and $D^\tau_{igt} = \mathbb{1}[t - g = \tau]$ is an indicator for event time, measured in decades relative to the treatment cohort. α_{ig} are tract-by-treatment-cohort fixed effects, which absorb time-invariant differences between recipient and control tracts, while allowing the same control tract to enter more than one treatment cohort stack. λ_{gt} represent treatment cohort-by-decade fixed effects, so that identification comes specifically from comparing differences between recipient and matched control neighborhoods in each decade within each treatment cohort.

I omit $\tau = -1$, the decade immediately preceding public housing construction, so that each β_τ measures the differential change in the recipient neighborhoods relative to the reweighted control group since that pre-treatment decade.

The primary outcomes I study are the Black population share, population by race, median income and rent, and homeownership rates. In the main event studies, I estimate the Black population share and homeownership rates in levels, and median income and rent in natural logs, using weighted least squares.

For population by race, some observations are zero, making their logarithms undefined. Log-like transformations such as the inverse hyperbolic sine transformation accommodate zeros, but their estimated treatment effects can depend on the units of measurement and do not generally allow for a percentage interpretation (Chen and Roth 2024; Fitzgerald et al. 2026). I therefore estimate effects on population by race using weighted least squares in levels and Poisson pseudo-maximum likelihood (PPML), both using the matching weights described above. For PPML, I specify the conditional mean as:

$$\mu_{igt} = \exp\left[\alpha_{ig} + \lambda_{gt} + \sum_{\tau \neq -1} \beta_\tau \left(D^\tau_{igt} \times R_{ig}\right)\right] \tag{3}$$

where μ_{igt} denotes the conditional mean population. The main population figures report PPML estimates as percentage changes: $100 \times [\exp(\beta_\tau) - 1]$.

Identification requires that, absent public housing construction, recipient neighborhoods and their matched, reweighted controls would have followed parallel trends within each treatment cohort. For the main event studies, I adjust standard errors for spatial correlation following Conley (1999), using a 2-kilometer cutoff.

6.1.2 Nearby Neighborhoods

Public housing construction may also affect neighborhoods that do not themselves contain a project. Indeed, many of the most pressing debates around mid-century public housing concern the extent to which its effects contributed to broader urban change, such as precipitating white flight (Jackson 1985). I define nearby neighborhoods as census tracts that do not contain public housing but lie within one kilometer of a public housing project, as described in Section 4. I estimate the effects on these neighborhoods using a similar matching strategy.

In particular, I consider nearby tracts to be the "treated" tracts and apply the same matching and reweighting procedure used for recipient tracts, described above. Within each treatment cohort and donor-geography specification, I estimate a separate propensity score model for nearby-tract status using each tract's pre-treatment characteristics. I then select the three nearest controls from the common donor pool, pool the distinct selected controls, re-estimate the propensity score model on the selected sample, and reweight the controls to match the nearby tracts' covariate distribution. Finally, I estimate the same stacked difference-in-differences specification described above. Table 6 reports covariate balance for nearby neighborhoods and their matched controls.

6.2 Effects on Recipient Neighborhoods

Public housing construction substantially reshaped the racial and socioeconomic composition of recipient neighborhoods. Throughout this section, the figures report estimates using all three donor pools, but I report magnitudes from the national donor pool specification.

Figure 7 shows the matched event study estimates for the neighborhood Black population share. I find large effects of public housing construction on the subsequent share of Black residents in recipient neighborhoods. At the first census following construction, the estimated increase is 2.4 percentage points relative to the matched control neighborhoods. The effect grows over time, reaching 9.4 percentage points three decades later, or 32% of the recipient tracts' Black share of 29.5% in the census before construction.The results are similar across donor pools, and the estimated pre-period coefficients are generally flat. To examine the population changes underlying this shift, Figure 8 shows the results on population by race, estimated using both PPML and weighted least squares in levels. These results show that the rising Black share was mainly driven by Black population growth: three decades after construction, the Black population increased by 16.0% (PPML) or roughly 305 residents (WLS) relative to national controls. The estimates on

white population are negative in the long run, though they are usually statistically insignificant.

These population changes include the public housing residents themselves. Figure 9 shows effects on the private population, defined as census-reported population minus the estimated public housing population. Three decades after construction, the estimated changes in private population are -20.4% in total, -23.2% for Black residents, and -14.6% for white residents (PPML), or -924, -354, and -262 residents (WLS). The rise in Black population therefore comes from project tenants, while the private Black population fell soon after construction and the private white population declined gradually.[13]

I also find that public housing construction produced persistent changes in the economic composition of recipient neighborhoods. Figure 10 shows the effects of public housing on median rent, income, and homeownership rates, showing broad declines in all three relative to their matched controls. I find median income and homeownership rates fall immediately after construction, while median rents do not change initially but decline over the longer run. Three decades after construction, the estimated changes are -15.6% for median income, -16.2% for median rent, and -5.8 percentage points for homeownership.[14]

6.3 Effects on Nearby Neighborhoods

Figure 11 shows the effects on the Black population share in nearby neighborhoods. Three decades after construction, the share is 6.0 percentage points higher in the national donor specification, or 39% of the nearby tracts' Black share of 15.5% in the census before construction. Unlike in the recipient neighborhoods, Figure 12 shows that this shift comes from white population outflows. Three decades after construction, the estimated changes are -14.3% for white population and -12.5% for total population (PPML), and the estimated change in Black population (-7.0%) is not statistically significant.

Figure 13 shows the effects of public housing construction on median rent and income in the nearby neighborhoods. I find declines in median income in nearby neighborhoods, with an estimated immediate change of -2.0% with national donors and similar estimates under the other pools. Three decades after construction, the point estimates for median income remain negative across all three donor pools, though none is statistically significant. These results suggest either that lower-income individuals sorted into these nearby

[13]Project residents are measured in the 1970s and held at that count from the first census after opening through 2010, with racial composition varying by census year (Appendix A.3). Errors in these counts pass directly into the private series.

[14]The figures display log coefficients for median income and rent. I report their implied percentage changes as $100[\exp(\beta) - 1]$.

neighborhoods or that public housing led to income declines, but neither interpretation suggests dramatic economic deterioration driven by public housing, on average.

By contrast, the estimated effects on nearby rents are close to zero and imprecise, even though nearby neighborhoods lost population. Two features of these neighborhoods can explain why rents did not fall. First, the housing stock shrank along with the population, with an estimated change in nearby housing units of -100 relative to controls three decades after construction. Second, in segregated housing markets, Black households often had to pay more than white households for housing of similar quality. At midcentury, Black renters paid a premium for comparable housing in segregated cities (Cutler, Glaeser, and Vigdor 1999), and rents often rose substantially on blocks transitioning from white to Black occupancy (Akbar et al. 2022).

6.4 Robustness Checks

I next check how sensitive the main estimates are to the number of matched controls, spatial inference, project location, and functional form. Figures A5 and A6 report estimates with one, three, or five matched controls under each donor-pool rule. Table A4 compares Conley standard errors using 2 km and 5 km distance cutoffs for effects three decades after construction, holding samples, matching weights, and fixed effects constant. Table A5 restricts the sample to tracts whose projects are located from geocoded building addresses, and the estimates three decades after construction are similar.

Another potential concern is that the functional form used for outcomes could affect the interpretation of results. Recent work by Chen and Roth (2024) highlights that log-like transformations in difference-in-differences designs can produce estimates that are sensitive to the choice of units and may conflate extensive and intensive margin effects. McConnell (2024) further shows that when treated and control groups have different baseline means, log specifications can yield misleading estimates and even produce opposite signs relative to levels specifications. The population analysis therefore uses PPML, which measures proportional changes and accommodates zero counts without transforming the outcome, alongside weighted least squares in levels, which measures changes in numbers of residents. Income and rent are measured in natural logs.

A more fundamental concern is that residential sorting induced by public housing could directly affect control tracts in the same metropolitan area, violating the stable unit treatment value assumption (SUTVA). For instance, households displaced or deterred by a new project might relocate to observationally similar control neighborhoods within the same metropolitan area.

If the results are driven by SUTVA violations, we might expect offsetting movements between the control tracts and recipient tracts. For example, we might see the Black population share in matched controls decline as residents move to recipient neighborhoods, which could drive the event-study results. Figure A2 plots raw Black shares for recipient and nearby neighborhoods and their weighted controls under each of the three pooled donor designs over event time. These descriptive paths help assess this mechanism, although they do not rule out spillovers to controls. Under every donor pool, the Black share of the weighted controls rises after construction rather than falls, for controls matched to recipient tracts and to nearby tracts alike. The estimated effects therefore reflect faster increases in recipient and nearby neighborhoods, not declines among their controls.

Another concern is that unobserved shocks between census dates could correlate with both project site selection and neighborhood outcomes. For example, if a neighborhood experienced an influx of Black residents after the baseline census, this demographic shift could have influenced where public housing was sited. Because matching relies on baseline census characteristics, it would miss these changes, and the estimates could reflect pre-existing demographic trends rather than the causal effect of public housing. To assess this, I exploit the fact that projects open at different points within the decade. A project that opens shortly after a census is matched on nearly contemporaneous covariates, while one that opens just before the next census is matched on covariates up to nine years stale. I classify recipient tracts as early openers (5 to 10 years before the next census) or late openers (0 to 4 years before the next census), pooling the 1950, 1960, and 1970 cohorts, and estimate separate event studies for each group. The 1980 cohort is excluded because the project inventory ends in 1973, so every project in that cohort opened 7 to 10 years before the 1980 census, and the cohort has no late openers (244 of 1,186 recipient tracts). If changes between census dates confound the estimates, effects may differ for late openers, whose matching characteristics are measured further before construction. Figure A3 compares the estimated effects three decades after construction for early and late openers. With national donors, the effects on the Black population share, median income, and median rent are similar for the two groups, and none of these contrasts is statistically significant at the 5% level. Total and white population grow more in the tracts of early openers, and both contrasts are significant.

Finally, I ask whether the main results are driven by particular subsets of projects: those in New York City, those in neighborhoods also targeted by urban renewal, and those later selected for HOPE VI revitalization. For categories that are also defined for control tracts (New York City and urban renewal), I classify the matched controls in the same way, rescale the control weights within each category and treatment cohort to sum to the num-

ber of recipient tracts in that cell, and include category-by-treatment-cohort-by-decade fixed effects so that recipient tracts are compared with matched controls in the same category. For HOPE VI selection, which is defined only for recipient tracts, both groups share the matched controls and weights from Section 6.1.1.

The New York City Housing Authority (NYCHA), the nation's largest housing authority, is widely regarded as better managed and more successful than its peers (Bloom 2008; Hunt 2018), and accounts for a substantial share of the recipient tracts in my sample. Figure 14 shows results separately for New York City and for other cities. The effects are not driven by New York City: recipient neighborhoods elsewhere show increases in Black population share and declines in income and rents of the same character, and with national donors the Black-share increases are not distinguishable between the two groups. The long-run rent decline is larger in New York City, and the contrast is statistically significant at the 5% level, whereas the income contrast is estimated imprecisely. The neighborhood effects of public housing therefore do not neatly correspond to the perceived success of local housing authorities.

I next ask whether the results are driven by urban renewal. Section 5 showed that recipient neighborhoods were more likely to be targeted by urban renewal, so the estimates could conflate the two programs. Figure 15 presents the results, separately estimating Equation 2 for neighborhoods that were urban-renewal tracts and those that were not. The effects are not driven by urban renewal: using national donors, recipient tracts in both groups show statistically significant long-run increases in Black population share and declines in income and rents, and none of the contrasts between the two groups is statistically significant at the 5% level for the displayed outcomes. Because urban-renewal status can coincide with or follow public housing construction, these comparisons do not separately identify the effects of the two programs, but the persistence of the effects outside urban-renewal tracts shows that the main pattern is not confined to areas where the programs overlapped.

Last, I ask whether the results are driven by the projects that later proved most distressed. I link projects to the HOPE VI award roster described in Section 4. Figure 16 compares recipient neighborhoods whose projects were later selected for HOPE VI ($N = 195$) with those whose projects were not ($N = 983$). The effects are not confined to the projects later selected for HOPE VI: Black population shares rise by the same amount in both groups, and income and rents fall significantly in both. Neighborhoods with later-HOPE VI projects do show larger long-run declines in income and rents, consistent with the distress that led to their selection.

7 Mechanisms and Heterogeneity

Next, I explore heterogeneity in the effects of public housing construction across several dimensions. In particular, I examine how the effects vary with construction cohort and project characteristics. To estimate heterogeneous effects, I interact treatment with the category of interest. Because each category in this section is a characteristic of the projects, all categories share the matched controls and weights from Section 6.1.1. Where the text quotes percentage changes in population counts, they are PPML estimates on the same samples; the figures show weighted least-squares effects in residents.

7.1 Construction Cohort

I first test whether the effects varied with when projects were built, estimating a separate effect for each construction cohort: the 1950 cohort (first qualifying project completed April 1940 to March 1950, $N = 277$), the 1960 cohort (April 1950 to March 1960, $N = 296$), the 1970 cohort (April 1960 to March 1970, $N = 369$), and the 1980 cohort (April 1970 to 1973, $N = 244$). Each cohort keeps its own matched controls and weights, and the four treatment paths are estimated jointly. Figure 17 shows the estimates.

The long-run effects are larger for the earlier cohorts. With national donors, three decades after construction the Black population share rises by 14.0 percentage points in the 1950 cohort against 5.4 in the 1970 cohort, and median income falls by 26.5% against 5.9%. The income declines of the 1950 and 1960 cohorts differ significantly from that of the 1970 cohort under every donor pool, while rent declines are similar across cohorts.

Earlier projects were also larger: recipient tracts in the 1950 and 1960 cohorts received 649 and 771 public housing units on average, against 465 and 196 in the 1970 and 1980 cohorts. Cohorts also differ in their cities, their projects, and the calendar periods over which the effects accumulate, so these comparisons do not identify the effect of building earlier.

7.2 Family versus Elderly Projects

Why did earlier projects have much larger neighborhood effects than those built in the 1960s and early 1970s? As discussed in Section 2, mounting criticism of large family projects led housing authorities in the 1960s to shift construction toward developments designated for elderly tenants (National Commission on Urban Problems 1968). To test whether this shift accounts for the cohort differences, I classify projects into three groups:

early family projects (opened 1940–1960), later family projects (opened 1960–1973), and later elderly projects (opened 1960–1973).

Figure 18 shows that the effects in both recipient and nearby neighborhoods are driven by family projects—and by early family projects in particular. Three decades after construction, early family projects increase the recipient Black population share by 13.1 percentage points and reduce median income by 24.4% (-0.280 log points). Later family projects also reduce recipient median income by 16.7% (-0.182 log points), whereas elderly projects built during the same 1960–1973 period have virtually zero effect on neighborhood median income and a smaller effect on racial composition than early family projects. Similarly, white population outflows in nearby neighborhoods are concentrated around family projects, while elderly developments induce no statistically significant white flight.

7.3 Project Size

Section 3 raised the physical scale of projects as one channel through which they could change their surroundings, and early projects were larger than later ones (Section 7.1). To test whether size accounts for the larger effects of early family projects, I split them (opened 1940–1960) at the median project size of 535 units for recipient tracts and 720 units for nearby tracts (Figure 19). Within early family projects, project size does not explain the racial turnover: below-median and above-median early family developments generate nearly identical increases in recipient Black population shares (12.9 and 13.2 percentage points), and neither the white outflows in nearby tracts (14.3% and 24.1%) nor the declines in recipient median income (22.1% and 26.6%) differ significantly between the two groups. Appendix C.2 reports effects by terciles of project size across all recipient tracts. Income declines are larger in the largest tercile, but the terciles largely sort project types: the smallest consists mostly of later elderly projects and the largest mostly of early family projects.

7.4 Project Tenant Racial Composition and Transition

If physical project size does not explain why early family projects drove neighborhood racial transition, what does? Using my longitudinal dataset of project-level tenant demographics (Appendix A.3), I examine the racial makeup of the projects themselves. By 1977, early family projects (opened 1940–1960) and later family projects (opened 1960–1973) housed predominantly Black tenants—averaging 70.6% and 80.4% Black occupancy, respectively—compared to just 24.0% in elderly projects opened in 1960–1973.

Even among early projects (opened 1940–1960), however, tenant racial composition followed three distinct paths between their initial decade of operation and the 1970s (Figure 20): (i) projects that remained under 50% Black (rising from 15.7% to 28.5% Black on average), (ii) projects that opened as predominantly white or integrated but transitioned to majority-Black occupancy by the 1970s (increasing from 16.4% to 79.0% Black on average), and (iii) projects that were already majority-Black at opening (averaging 85.7% Black initially and 91.7% in the 1970s).

Estimating separate effects across these three trajectory groups shows that the neighborhood changes closely track the racial composition of the projects' tenants (Figure 21). In recipient neighborhoods, increases in the Black population share and declines in median income are largest around projects that transitioned to majority-Black occupancy after opening: three decades after construction, these tracts experienced a 28.8 percentage point increase in the Black population share and a 36.7% decline in median income, compared to 9.1 percentage points and a 15.8% decline where projects stayed under half Black, and 6.8 percentage points and a 19.1% decline where projects were already majority Black at opening.

White population outflows, however, occur in both recipient and surrounding neighborhoods around projects that opened majority Black and those that became majority Black. Three decades after construction, the white population fell by 40.8% in recipient tracts and 23.8% in nearby tracts around projects that became majority Black, and by 55.1% in recipient tracts and 44.2% in nearby tracts around projects that were majority Black at opening.[15] By contrast, around early projects that remained under half Black, the white population of nearby tracts grew by 29.2% and their Black population share did not rise. The white population of the recipient tracts in that group rose by 40.4%, an estimate too imprecise to distinguish from zero (95% interval -3.1 to 103.3 percent).

These comparisons show that the effects of public housing are concentrated in a particular set of projects: early family developments opened between 1940 and 1960. Project size does not explain why these early developments reshaped neighborhoods, since projects above and below the median size raised the recipient Black population share by similar amounts. Instead, the effects follow the racial makeup of the projects' tenants: in recipient neighborhoods, the Black population share rose and median income fell most where a project's tenants became majority Black after opening, while white residents left both re-

[15] In recipient tracts of projects that were majority Black at opening, the white-population effect two decades before construction, relative to the decade before, is 30.9% [15.0, 49.1]. White population in these tracts was already falling relative to matched controls before the projects opened, so part of the post-construction decline may continue that trend.

cipient and nearby neighborhoods only where a project was or became majority Black. Because project cohort, family designation, and tenant racial composition overlap—and because a project's tenants changed alongside its neighborhood—these comparisons do not isolate a single causal channel. They also leave open whether white residents responded directly to the race of the projects' tenants or to correlated local conditions, such as crime, that may have changed as the projects did, the sorting and externality channels of Section 3.

8 Public Housing and Long-Run Economic Opportunity

I next examine the adult outcomes of children who grew up in recipient and nearby neighborhoods. This exercise connects the neighborhood changes documented above to later economic opportunity. In particular, I ask whether children living in recipient and nearby neighborhoods had different levels of upward mobility and incarceration rates than those in their matched controls. To do so, I use data on upward mobility and adult incarceration rates for low-income children born between 1978 and 1983 from Chetty, Friedman, et al. (2018), as described in Section 4. I merge these outcomes into the matched samples from Section 6, retaining the selected controls and weights from each neighborhood design. Appendix Figure A7 maps upward mobility around Robert Taylor Homes in Chicago, illustrating differences in adult income ranks across nearby census tracts.

I estimate the following cross-sectional model separately for recipient tracts and nearby tracts:

$$Y_i = \beta R_{ic} + \mu_c + \epsilon_{ic} \tag{4}$$

where Y_i denotes the Opportunity Atlas outcome for tract i, and R_{ic} indicates recipient or nearby status. c represents the comparison set, a public housing construction cohort, with selected donors pooled within each cohort, and μ_c is a cohort fixed effect. I estimate Equation 4 using weighted least squares with the same weights as described in Section 6.1. Standard errors are again adjusted for spatial correlation within a 2-kilometer radius following Conley (1999).

The baseline specification includes no additional control variables. I also estimate a version in which I control for 1980 tract-level racial and economic characteristics: in particular, Black population share, median income, and unemployment rate. These characteristics are affected by public housing itself and are therefore "bad controls," so the adjusted estimates describe conditional associations. Their attenuation can indicate a relationship with the previously documented demographic and socioeconomic changes, but does not

identify a mediated share or isolate an independent effect of public housing.

Table 7 presents the results by donor geography. Panel A reports adult household income rank and Panel B reports incarceration, and each pair of columns reports specifications with and without 1980 neighborhood controls. I find that recipient neighborhoods had notably lower rates of upward mobility along with higher adult incarceration rates. In the primary specification, recipient neighborhoods have lower mean income ranks by 2.5, 2.6, and 1.9 rank points relative to weighted donors from the same Census division, the same Census region, and the national pool, respectively. Their incarceration rates are also higher by 0.8, 0.9, and 0.7 percentage points. These differences are statistically significant at the 5% level under all three donor pools. However, after controlling for racial and economic composition, the income-rank gaps disappear and the incarceration gaps shrink by more than half, consistent with these differences being related to the demographic and economic characteristics of recipient neighborhoods.

Nearby tracts also exhibit lower adult income ranks and higher incarceration rates, with gaps about half as large as those in recipient tracts. The income-rank differences are statistically significant at the 5% level under all three donor pools, and the incarceration differences are significant with same-region and national donors. These differences shrink and become statistically insignificant after adding controls.

Ultimately, these results suggest that recipient neighborhoods were lower-opportunity neighborhoods for children who grew up there, with lower rates of upward mobility and higher rates of incarceration. Still, the fact that these differences shrink substantially after conditioning on neighborhood racial and economic characteristics in 1980 suggests that this disadvantage is related to these broader neighborhood characteristics. Together with the evidence above that public housing changed neighborhood composition, this pattern is consistent with demographic and economic changes contributing to long-run opportunity differences. These conditional comparisons cannot establish that channel or rule out other effects of public housing.

9 Conclusion

In this paper, I assemble a new national dataset to evaluate how one of the largest urban policy interventions in U.S. history reshaped American neighborhoods. I first show that project placement was far from neutral: consistent with both the program's slum-clearance mandate and local resistance from white communities, public housing was systematically sited in poorer neighborhoods with higher initial shares of Black residents. I then demonstrate that public housing construction did not merely reflect pre-existing

differences, but actively accelerated demographic and economic divergence. Over the three decades following construction, recipient neighborhoods experienced large, persistent increases in their Black population shares alongside steep declines in median incomes, rents, and homeownership rates. Surrounding neighborhoods within one kilometer of public housing also experienced racial transition driven by substantial white population outflows, though with limited evidence of broader economic deterioration.

Crucially, these neighborhood effects were not uniform across the public housing stock. The long-run effects were concentrated among earlier-generation family developments: recipient Black population shares rose and incomes fell most where projects transitioned to majority-Black occupancy after opening, while white residents left both recipient and surrounding neighborhoods around projects that opened with or transitioned to majority-Black tenant populations. By contrast, post-1960 elderly developments and early projects whose tenants remained under half Black generated virtually no white flight in surrounding areas. These neighborhood transformations also carried implications for intergenerational mobility: low-income children who grew up in recipient and nearby tracts experienced lower adult income ranks and higher incarceration rates than those in matched control tracts, gaps that shrink by more than half once 1980 neighborhood composition is held fixed.

Taken together, these findings show that the critical indictment of mid-century American public housing is at least partially correct, but requires important qualifications. On the one hand, federal public housing both reflected and reinforced urban inequality, concentrating poverty and entrenching racial segregation in the neighborhoods where the projects were built (Rothstein 2017; Trounstine 2018). On the other hand, the program did not generally cause broad deterioration across surrounding areas, and its sharpest neighborhood spillovers were confined to projects whose tenants were or became majority Black. These patterns also help rationalize modern shifts in low-income housing policy toward mixed-income developments designed to integrate into existing neighborhoods.

Still, finding long-run effects on neighborhoods does not imply that public housing harmed the tenants that it housed. Public housing may have offered low-income families safer, higher-quality, and more affordable housing than their available private-market alternatives, as evidenced by the fact that public housing has often been oversubscribed. A full assessment of the program must account for its effects on residents themselves. In ongoing work, I am taking up that question by linking individuals to the public housing projects in which they lived.

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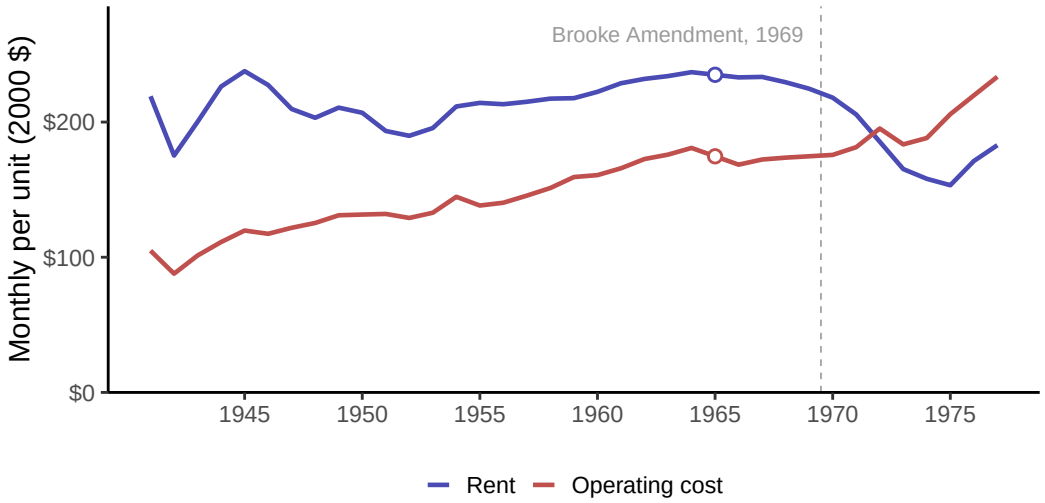
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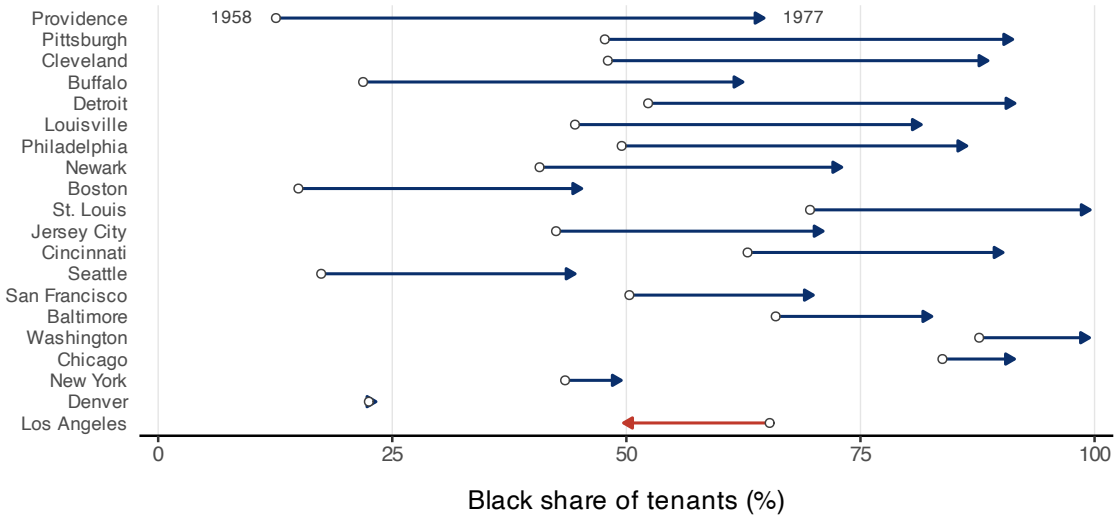
Figures and Tables

Figure 1: Rent and Operating Cost per Unit in Public Housing, 1941–1977



Note: Rent is dwelling rental income net of vacancy loss. Operating cost is management, operating services, utilities, and repairs and maintenance through fiscal 1958, and administration, utilities, and ordinary maintenance and operation from fiscal 1959. Both are per dwelling unit per month, in 2000 dollars (CPI-U). Fiscal years end June 30. The series covers the original 1937-Act projects through fiscal 1954 and all locally owned projects eligible for annual contributions from fiscal 1955. Neither source prints fiscal 1965, and the hollow point lies on the straight line between 1964 and 1966. Data come from the Housing and Home Finance Agency's annual reports (1947–1964) and HUD's Statistical Yearbooks (1966–1977).

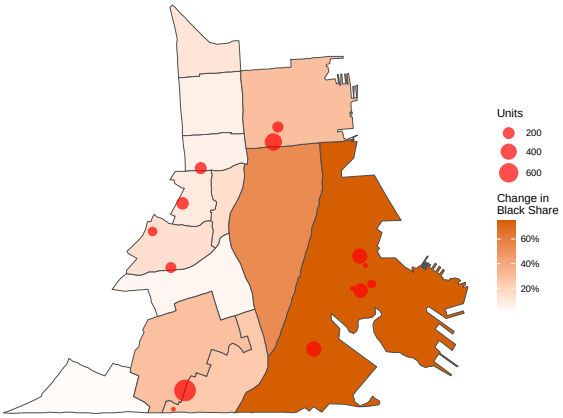
Figure 2: Black Share of Public Housing Tenants by City, 1958 and 1977



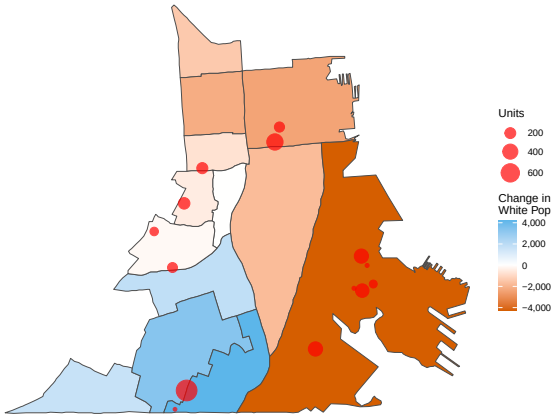
Note: The figure shows the 20 cities with the most units in federal projects completed before 1960, ordered by the change. Open dots give the share of occupied units occupied by Black families on March 31, 1958, as printed in the Public Housing Administration's *Trends Toward Open Occupancy in Low-Rent Housing Programs*. Arrowheads give the unit-weighted Black share of tenants in the same city's federal projects completed before 1960, from the 1977 Picture of Subsidized Households, except in New York (New York City Housing Authority family counts, December 1971) and Baltimore (Housing Authority of Baltimore City figures, December 1970). Red marks the city where the share fell. Appendix A.3 describes the project shares.

Figure 3: Motivating Example: Hunters Point, San Francisco

(a) Change in Black Population Share

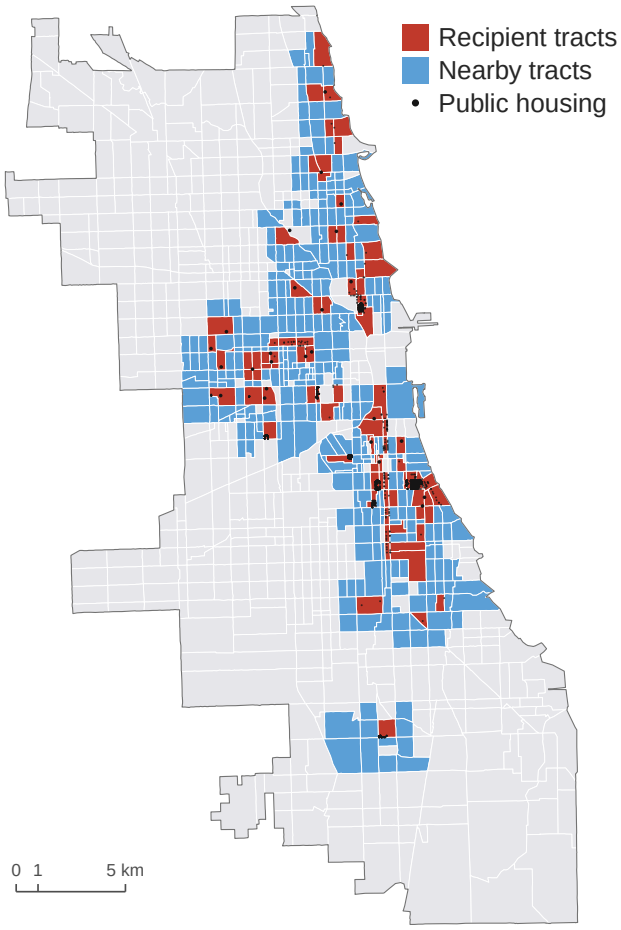


(b) Change in White Population



Note: These maps show demographic changes from 1940 to 1970 in the census tract containing the Hunters Point public housing projects and the surrounding tracts within 2 kilometers (tracts with fewer than 100 residents in 1950 are omitted). Public housing first opened at Hunters Point in 1942–1944; the permanent projects in the inventory, Hunters Point, Hunters View, and Alice Griffith Garden Homes, were completed between 1953 and 1962. Panel (a) shows the change in Black population share; panel (b) shows the change in white population. Red dots mark every public housing project in the inventory located in the mapped tracts, with area proportional to the project's units. See Section 2 for historical context.

Figure 4: Public Housing Projects and Nearby Areas: Chicago



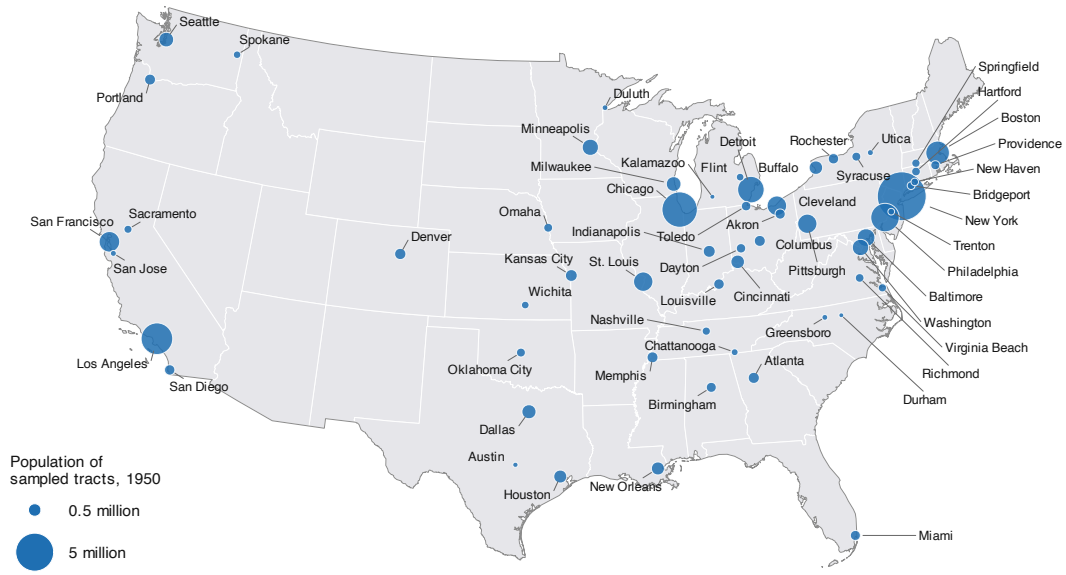
Note: This map shows census tracts in Chicago containing public housing projects built between 1941 and 1973 (red, N=92 tracts) and nearby areas (blue). Black dots mark public housing buildings; projects located only through project-level coordinates are shown as a single point. Spillover areas are defined as census tracts that do not themselves contain a project but lie within 1 kilometer of a qualifying project's buildings. See Section 4 for sample construction details.

Table 1: Sample Attrition

Sample restriction	Tracts	1940 pop (mill.)	CBSAs	Recipient tracts	Projects	Units
Original sample	12,062	38.2	60	1,471	1,521	518,056
Observed from 20 years before opening	11,880	37.2	59	1,375	1,410	498,872
Complete variables	11,490	36.9	59	1,352	1,400	493,757
Exclude treated ≤1940	11,416	36.5	59	1,278	1,324	460,343
Drop sparsely populated tracts	9,759	35.3	59	1,186	1,213	438,003

Notes: Restrictions are applied sequentially. "Observed from 20 years before opening" requires an observation in every census decade from 20 years before the cohort through 2010. "Complete variables" additionally requires nonmissing population, Black population share, median family income, median contract rent, labor-force participation, and unemployment in each of those decades. Tracts whose first qualifying project was completed by the April 1940 census are excluded to retain two pre-treatment observations. The final step excludes tracts with fewer than 250 residents or more than 8 people per household in 1940 and geographic outliers exceeding 10 km^2. Recipient tracts received public housing; projects and units refer to public housing. Population is reported in millions. The final sample contains approximately 27% of the 1940 U.S. population.

Figure 5: Geographic Coverage of CBSAs in Analysis Sample



Note: This map shows the 59 Core-Based Statistical Areas (CBSAs) included in the balanced panel analysis, containing 9,759 census tracts and 1,186 recipient tracts. Each circle is centered on a CBSA, with area proportional to the 1950 population of its sampled tracts. Table 1 lists the full set of sample restrictions.

Table 2: Comparison of Recipient and Non-Recipient Neighborhoods (1940 Baseline)

Variable	Non-recipient	Recipient	Std. Diff.
N (Tracts)	7,488	1,118	
Total Population (asinh)	8.74 (0.75)	9.00 (0.73)	0.352***
Black Share	0.06 (0.16)	0.19 (0.30)	0.570***
Black Population (asinh)	2.97 (2.57)	4.94 (3.08)	0.694***
Median Income (log)	3.02 (0.32)	2.75 (0.33)	0.819***
Median Rent (log)	5.90 (0.41)	5.65 (0.40)	0.613***
Unemployment Rate	0.11 (0.06)	0.15 (0.06)	0.559***
Labor Force Participation Rate	0.55 (0.05)	0.56 (0.05)	0.145***
Distance from CBD (asinh)	7.98 (3.00)	6.98 (3.47)	0.308***
Share Needing Major Repairs	0.08 (0.10)	0.14 (0.14)	0.458***

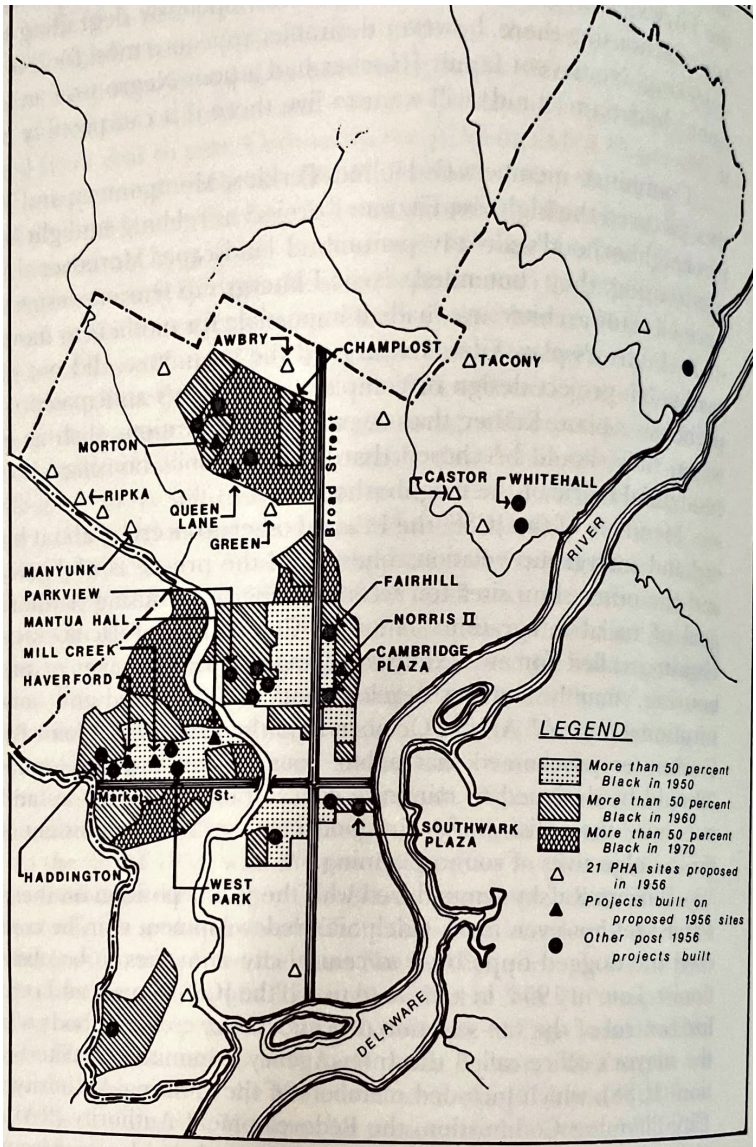
Notes: This table compares 1940 baseline characteristics between census tracts that eventually received public housing projects between April 1940 and 1973 and tracts that did not. Sample sizes are reported in the first row. Log denotes the natural logarithm; asinh denotes the inverse hyperbolic sine transformation. Standard deviations are shown in parentheses. The Std. Diff. column reports standardized mean differences (difference in means divided by pooled standard deviation). Statistical significance from two-sample t-tests: *p<0.1, **p<0.05, ***p<0.01. See Section 5 for analysis of site selection patterns.

Table 3: Public Housing Placement and 1940 Neighborhood Characteristics

	Received public housing, 1941–1973			
	(1)	(2)	(3)	(4)
Black population share	0.254***	0.242***	0.227***	0.234***
	(0.045)	(0.044)	(0.046)	(0.045)
Total population (log)	0.032***	0.027***	0.027***	0.026***
	(0.008)	(0.009)	(0.009)	(0.008)
Median family income (log)	−0.169***	−0.127***	−0.128***	−0.110***
	(0.031)	(0.037)	(0.038)	(0.035)
Median contract rent (log)	−0.030	−0.011	0.004	−0.010
	(0.023)	(0.028)	(0.029)	(0.028)
High-school graduate share		0.123**	0.112**	0.098*
		(0.053)	(0.054)	(0.052)
Unemployment rate		0.236*	0.244*	0.216*
		(0.129)	(0.130)	(0.127)
Labor-force participation rate		−0.335**	−0.312**	−0.311**
		(0.144)	(0.147)	(0.144)
Homeownership rate		−0.184***	−0.177***	−0.178***
		(0.036)	(0.036)	(0.036)
Share under age 18		0.517***	0.491***	0.547***
		(0.146)	(0.148)	(0.142)
Share age 65 and over		−0.030	−0.006	−0.010
		(0.229)	(0.234)	(0.226)
Redlined (HOLC grade D)		0.025	0.028	0.022
		(0.017)	(0.018)	(0.018)
Distance from CBD (asinh)		0.002	0.003	0.002
		(0.002)	(0.002)	(0.002)
CBD tract		−0.010	−0.022	−0.016
		(0.026)	(0.027)	(0.027)
Share needing major repairs			0.086*	
			(0.050)	
Distance to highway (asinh)				0.007
				(0.007)
Urban renewal area				0.069***
				(0.023)
Observations	8,606	8,599	8,196	8,599
R^2	0.130	0.142	0.138	0.145
Adj. R^2	0.125	0.136	0.132	0.139
CBSA fixed effects	Yes	Yes	Yes	Yes

Notes: This table reports linear probability model estimates of public housing placement on 1940 neighborhood characteristics. The dependent variable equals 1 if a tract received a project between April 1940 and 1973 (N=1,118) and 0 otherwise (N=7,488) in the Column (1) sample; its sample mean is 13%. Income, rent, and population enter in natural logs; distances use the inverse hyperbolic sine transformation. All columns include CBSA fixed effects using the fixed June 2003 delineations described in Section 4.2. Column (2) adds neighborhood controls; Column (3) adds housing disrepair, restricting the sample to tracts with repair data; Column (4) adds urban renewal and highway proximity to Column (2). Standard errors in parentheses allow spatial correlation within 2 kilometers following Conley (1999). Statistical significance: *p<0.1, **p<0.05, ***p<0.01.

Figure 6: Historical Map of Proposed and Actual Philadelphia Public Housing Sites, 1956



MAP 9: Post-1956 Philadelphia Housing Authority Projects and Racial Change, 1950–1970

Source: Adkins and Associates, Architects.

Note: This map shows the distribution of proposed and actual public housing sites in Philadelphia. Following the 1954 Housing Act, the Philadelphia Housing Authority proposed 21 projects across the city (triangles); 7 were built on the proposed sites (filled triangles) and 14 were canceled after opposition from white communities (open triangles). Circles mark other projects built after 1956. The 14 canceled sites fall in 12 census tracts, the proposed-but-not-built neighborhoods of Table 4. The proposed-but-not-built sites had systematically different baseline characteristics from actual sites (particularly lower Black population shares), illustrating both the political dynamics of site selection and why such sites would not serve as valid counterfactuals in a research design. Source: Bauman (1987), georeferenced by author.

Table 4: 1940 Neighborhood Characteristics: Proposed vs Actual Public Housing Locations in Philadelphia

Characteristic	All tracts	Built sites	Proposed-only sites	Absolute SMD: built vs proposed
Black population share	0.10	0.29	0.01	1.374***
	(0.19)	(0.28)	(0.02)	
Total population (asinh)	8.09	9.07	7.98	0.701**
	(2.14)	(1.30)	(1.78)	
Median family income (log)	3.00	2.81	2.96	0.481
	(0.31)	(0.30)	(0.29)	
Unemployment rate	0.15	0.21	0.14	0.931**
	(0.10)	(0.09)	(0.05)	
Median contract rent (log)	5.84	5.72	5.81	0.252
	(0.43)	(0.30)	(0.39)	
Distance from CBD (asinh)	8.75	8.41	9.76	0.952**
	(2.25)	(1.99)	(0.26)	
Redlined (HOLC grade D)	0.21	0.41	0.08	0.808**
	(0.40)	(0.50)	(0.29)	
Urban renewal area	0.17	0.37	0.17	0.466
	(0.37)	(0.49)	(0.39)	
Tracts	404	51	12	

Notes: This table reports 1940 characteristics of Philadelphia census tracts on 1950 boundaries. The citywide column includes all tracts in the descriptive sample, including built-site and proposed-only tracts; the paper's longitudinal sample restrictions are not imposed. Proposed-only tracts contain proposed-but-not-built sites identified from a 1956 Philadelphia Housing Authority map (Bauman 1987) and no built public housing sites. The first three columns report tract means, with standard deviations in parentheses beneath them; the final row reports tract counts. Statistics use nonmissing observations for each characteristic. Income and rent are in natural logs; population and distance from the central business district (CBD) use the inverse hyperbolic sine (asinh). The final column reports absolute standardized mean differences between built and proposed-only tracts. Stars indicate tests of equality of means for those two groups: *p<0.1, **p<0.05, ***p<0.01.

Table 5: Covariate Balance: Recipient Neighborhoods

Characteristic	Recipient tracts	Eligible controls (|SMD|)	Matched controls		
			Division (|SMD|)	Region (|SMD|)	National (|SMD|)
Matching covariates					
Total population (log)	8.29	8.16 (0.172)	8.24 (0.071)	8.23 (0.089)	8.25 (0.054)
Black population share	29.5	7.8 (0.749)	23.9 (0.165)	24.4 (0.150)	27.2 (0.066)
Black population (asinh)	5.86	3.24 (0.937)	5.36 (0.175)	5.39 (0.162)	5.64 (0.076)
Labor-force participation rate	57.2	56.9 (0.046)	57.3 (0.021)	57.2 (0.001)	57.0 (0.019)
Unemployment rate	11.2	6.4 (0.574)	10.4 (0.086)	10.8 (0.038)	10.5 (0.078)
Median family income (log)	3.12	3.50 (0.862)	3.18 (0.128)	3.17 (0.111)	3.14 (0.048)
Median contract rent (log)	5.68	5.98 (0.859)	5.71 (0.087)	5.70 (0.068)	5.68 (0.012)
Homeownership rate	31.4	51.9 (0.876)	34.1 (0.120)	33.6 (0.097)	34.1 (0.122)
Share under age 18	30.1	27.6 (0.336)	29.7 (0.057)	29.8 (0.041)	30.1 (0.005)
Distance from CBD (asinh)	7.03	8.50 (0.474)	7.17 (0.042)	7.26 (0.066)	7.05 (0.008)
Redlined (HOLC grade D)	28.6	7.8 (0.560)	23.6 (0.114)	24.9 (0.083)	25.9 (0.060)
Additional characteristics					
White population (asinh)	8.18	8.68 (0.381)	8.36 (0.119)	8.33 (0.099)	8.28 (0.063)
Housing needing major repair, 1940	13.5	6.5 (0.589)	11.9 (0.115)	11.8 (0.127)	12.4 (0.082)
High-school graduate share	28.9	42.6 (0.762)	30.0 (0.062)	29.3 (0.023)	29.4 (0.030)
Distance to highway (km)	1.83	2.38 (0.267)	1.88 (0.028)	1.90 (0.036)	1.78 (0.026)
Urban renewal area	24.0	2.5 (0.668)	11.0 (0.349)	9.5 (0.396)	9.7 (0.390)
Tract–cohort observations	1,186	17,808	1,979	2,087	2,156
With 1940 repair data	1,054	15,175	1,652	1,799	1,855

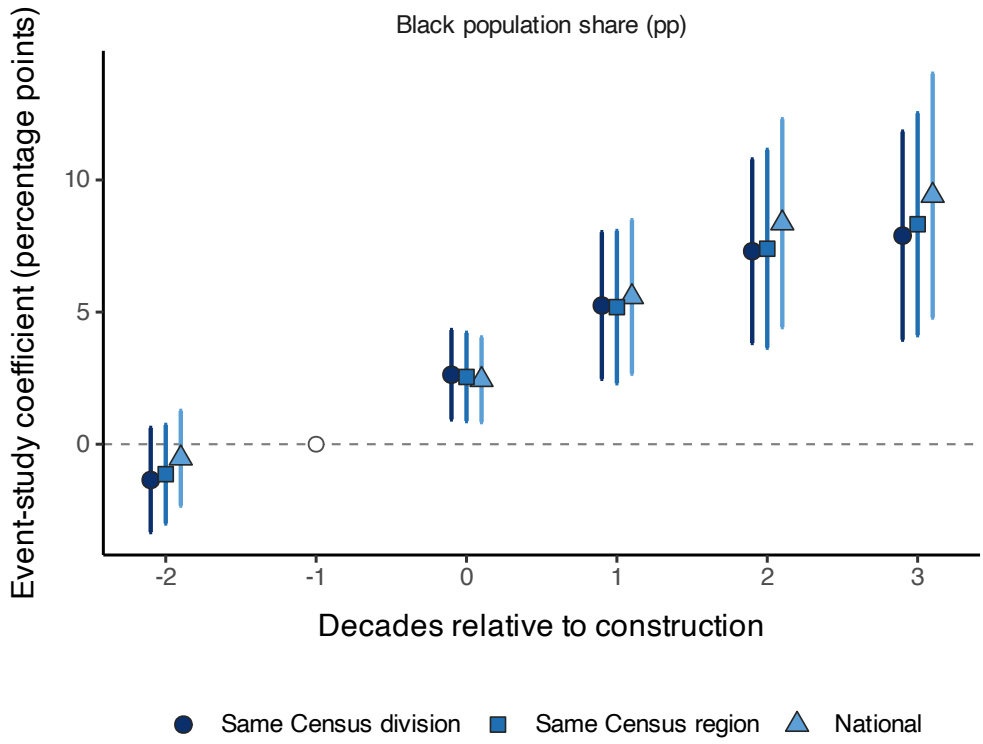
Notes: Means are shown with absolute standardized mean differences (SMDs) from recipients in parentheses. Eligible controls comprise the national donor pool before matching, including selected matches; recipient and nearby tracts are excluded. Matched controls use three nearest neighbors within the same Census division, Census region, or nationwide, followed by reweighting to improve covariate balance (Section 6.1.1). SMDs use each comparison's weighted pooled standard deviation. Census characteristics use the preceding census, except repair (1940 for every cohort). Shares and rates are percentages; counts are unweighted tract–cohort observations. Income is in thousands of 2000 dollars, rent in 2000 dollars, and distance from the central business district (CBD) in meters before transformation; asinh denotes the inverse hyperbolic sine. Redlined denotes at least 80% of tract area in HOLC grade D. Additional characteristics are excluded from matching. Repair is the share of units reporting condition that needed major repairs; its row excludes missing observations. Highways use the 1996 network; urban renewal denotes at least 5% overlap with historical project areas, which can coincide with or follow treatment.

Table 6: Covariate Balance: Nearby Neighborhoods

Characteristic	Nearby tracts	Eligible controls (|SMD|)	Matched controls		
			Division (|SMD|)	Region (|SMD|)	National (|SMD|)
Matching covariates					
Total population (log)	8.16	8.16	8.13	8.12	8.14
		(0.003)	(0.045)	(0.059)	(0.027)
Black population share	15.5	7.8	13.9	14.1	14.8
		(0.309)	(0.057)	(0.050)	(0.025)
Black population (asinh)	4.40	3.24	4.19	4.20	4.34
		(0.419)	(0.074)	(0.069)	(0.024)
Labor-force participation rate	58.1	56.9	58.0	58.0	57.9
		(0.204)	(0.004)	(0.019)	(0.031)
Unemployment rate	10.9	6.4	10.6	10.9	10.6
		(0.568)	(0.026)	(0.000)	(0.028)
Median family income (log)	3.20	3.50	3.22	3.22	3.20
		(0.689)	(0.045)	(0.043)	(0.004)
Median contract rent (log)	5.79	5.98	5.80	5.80	5.78
		(0.549)	(0.033)	(0.027)	(0.017)
Homeownership rate	34.4	51.9	35.7	35.6	35.6
		(0.743)	(0.056)	(0.055)	(0.052)
Share under age 18	27.1	27.6	27.3	27.3	27.1
		(0.069)	(0.014)	(0.022)	(0.002)
Distance from CBD (asinh)	7.37	8.50	7.38	7.48	7.30
		(0.372)	(0.002)	(0.033)	(0.022)
Redlined (HOLC grade D)	19.8	7.8	18.1	18.8	19.2
		(0.354)	(0.044)	(0.025)	(0.015)
Additional characteristics					
White population (asinh)	8.46	8.68	8.51	8.49	8.50
		(0.199)	(0.041)	(0.031)	(0.037)
Housing needing major repair, 1940	9.2	6.5	9.2	9.2	9.2
		(0.263)	(0.002)	(0.004)	(0.001)
High-school graduate share	32.1	42.6	32.5	32.1	32.9
		(0.563)	(0.019)	(0.000)	(0.046)
Distance to highway (km)	1.90	2.38	1.99	1.99	1.91
		(0.231)	(0.047)	(0.046)	(0.006)
Urban renewal area	15.4	2.5	8.6	7.4	7.7
		(0.463)	(0.209)	(0.253)	(0.244)
Tract–cohort observations	3,503	17,808	5,316	5,463	5,648
With 1940 repair data	3,201	15,175	4,628	4,786	4,939

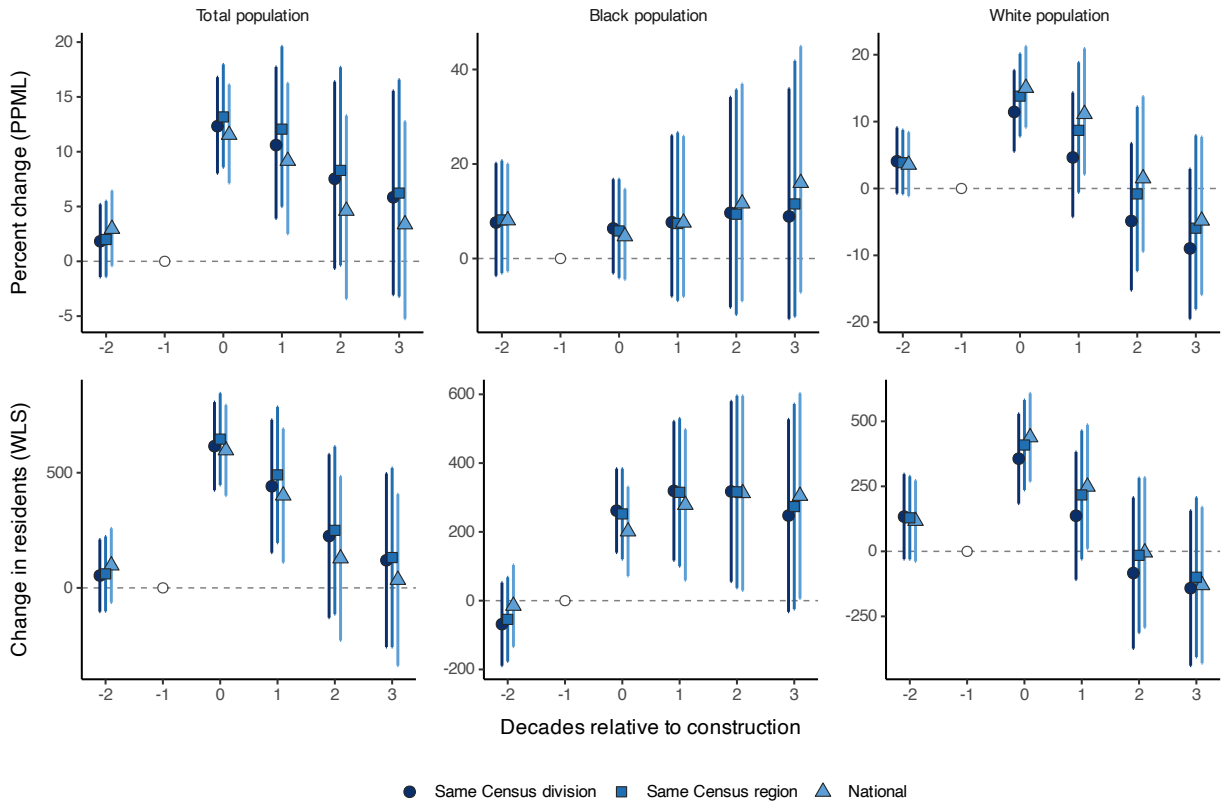
Notes: Means are shown with absolute standardized mean differences (SMDs) from nearby tracts in parentheses. Eligible controls comprise the national donor pool before matching, including selected matches; recipient and nearby tracts are excluded. Matched controls use three nearest neighbors within the same Census division, Census region, or nationwide, followed by reweighting to improve covariate balance (Section 6.1.1). SMDs use each comparison's weighted pooled standard deviation. Census characteristics use the preceding census, except repair (1940 for every cohort). Shares and rates are percentages; counts are unweighted tract–cohort observations. Income is in thousands of 2000 dollars, rent in 2000 dollars, and distance from the central business district (CBD) in meters before transformation; asinh denotes the inverse hyperbolic sine. Redlined denotes at least 80% of tract area in HOLC grade D. Additional characteristics are excluded from matching. Repair is the share of units reporting condition that needed major repairs; its row excludes missing observations. Highways use the 1996 network; urban renewal denotes at least 5% overlap with historical project areas, which can coincide with or follow treatment.

Figure 7: Effects on Black Population Share in Recipient Neighborhoods



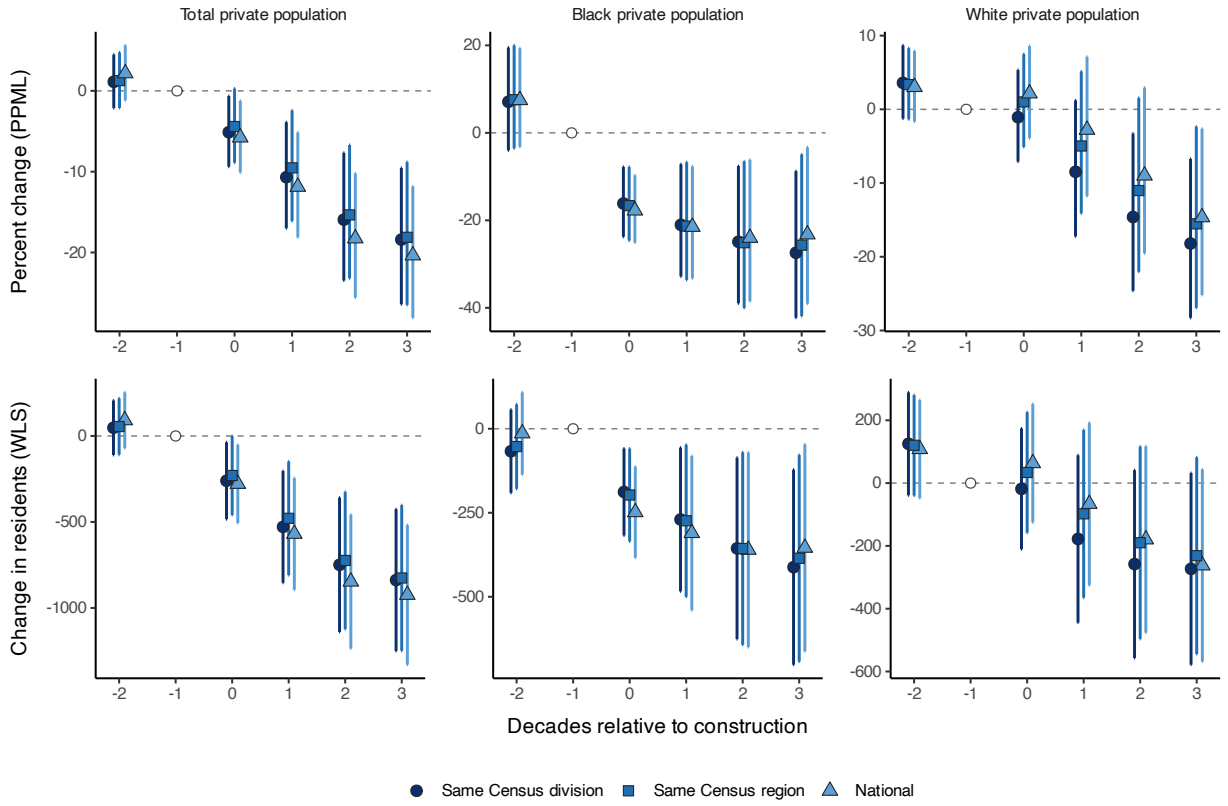
Note: Weighted least-squares event-study estimates of the effect of public housing construction on Black population share in recipient neighborhoods ($N = 1,186$), using Equation 2. Coefficients are percentage point changes. Three controls per recipient tract are matched on pre-construction characteristics, then pooled and reweighted. Same-division controls come from the recipient tract's Census division; same-region controls come from its Census region; national controls may come from any metropolitan area in the sample. Donor pools exclude recipient and nearby tracts. Bars show 95% confidence intervals with Conley standard errors (2 km cutoff). The reference period is $t = -1$, the decade before construction.

Figure 8: Population Effects in Recipient Neighborhoods: Percentages and Levels



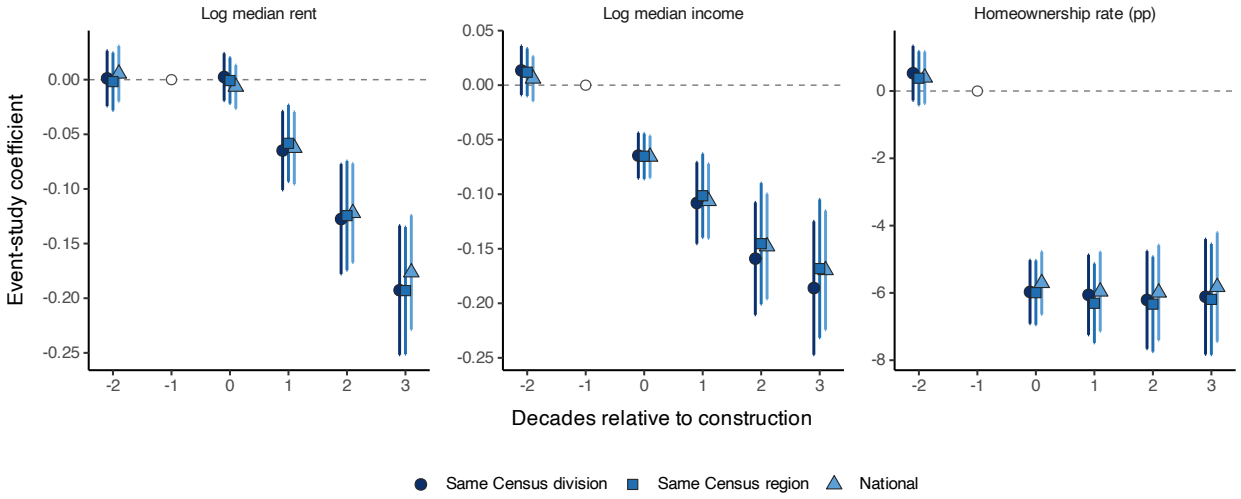
Note: Effects on total, Black, and white population in recipient neighborhoods ($N = 1,186$). The top row reports PPML percentage effects, $100[\exp(\beta) - 1]$, from Equation 3; the bottom row reports weighted least-squares effects in residents from Equation 2. Both start from the same matched samples and matching weights. Three controls per recipient tract are matched on pre-construction characteristics, then pooled and reweighted. Same-division controls come from the recipient tract's Census division; same-region controls come from its Census region; national controls may come from any metropolitan area in the sample. Donor pools exclude recipient and nearby tracts. Bars show 95% Conley confidence intervals (2 km cutoff); PPML interval endpoints are transformed to percentages. Open markers denote the omitted reference decade, $t = -1$.

Figure 9: Private Population Effects in Recipient Neighborhoods: Percentages and Levels



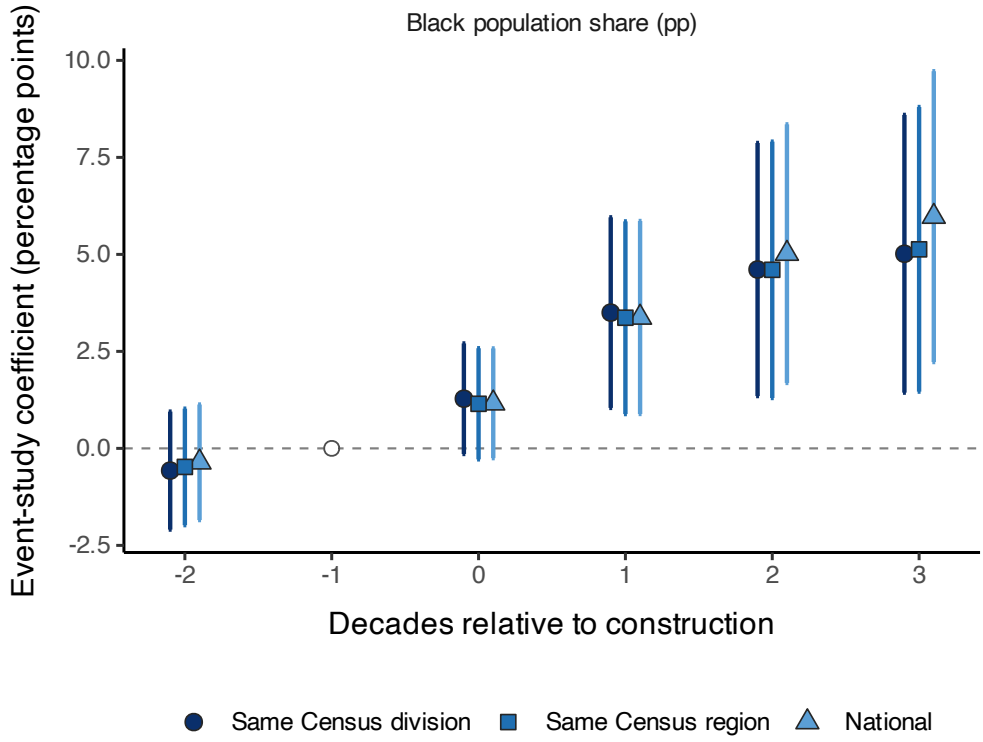
Note: Effects on the private (non-public-housing) population of recipient neighborhoods ($N = 1,186$), defined as census tract population minus the estimated number of public housing residents, with negative differences set to zero. Project residents are measured in the 1970s, set to zero before construction, and held at that count afterward; their racial composition varies by census year (Appendix A.3). The top row reports PPML percentage effects, $100[\exp(\beta) - 1]$, from Equation 3; the bottom row reports weighted least-squares effects in residents from Equation 2. Both start from the same matched samples and matching weights. Three controls per recipient tract are matched on pre-construction characteristics, then pooled and reweighted. Same-division controls come from the recipient tract's Census division; same-region controls come from its Census region; national controls may come from any metropolitan area in the sample. Donor pools exclude recipient and nearby tracts. Bars show 95% Conley confidence intervals (2 km cutoff); PPML interval endpoints are transformed to percentages. Open markers denote the omitted reference decade, $t = -1$.

Figure 10: Economic and Housing Effects in Recipient Neighborhoods



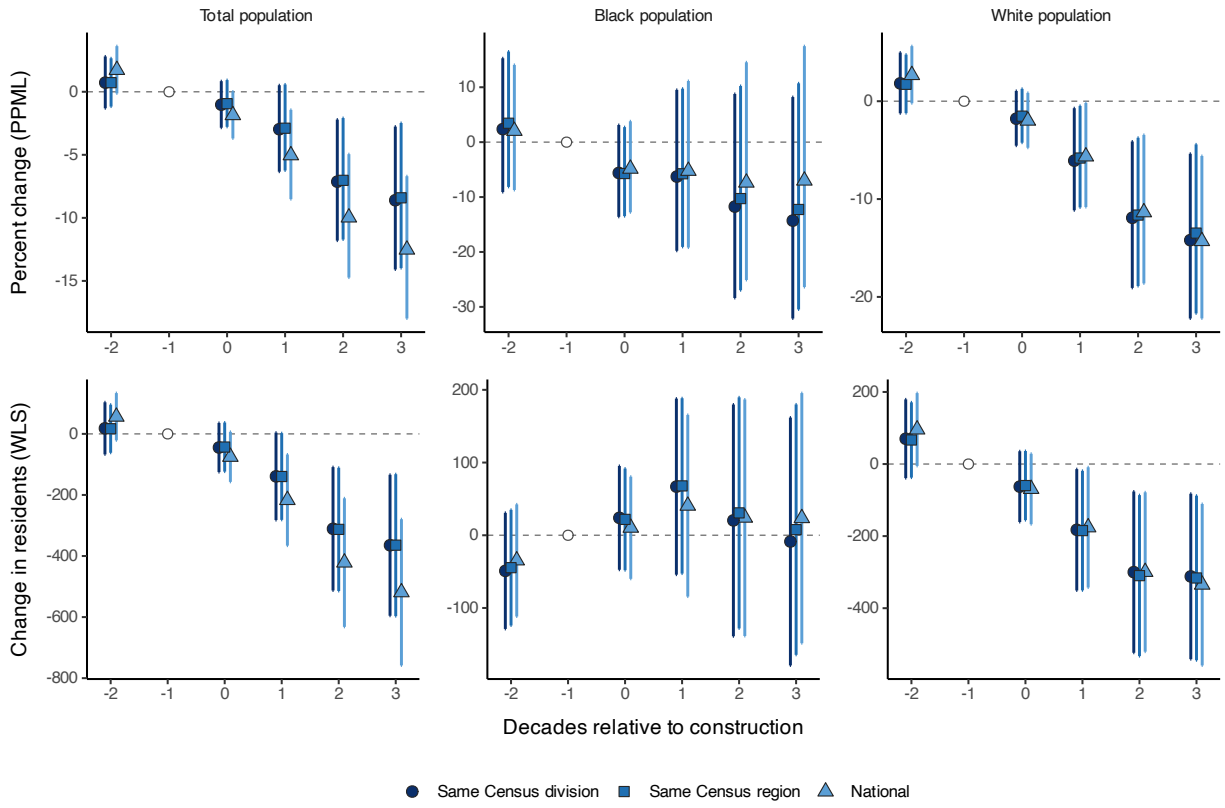
Note: Weighted least-squares event-study estimates for recipient neighborhoods ($N = 1,186$), using Equation 2. Outcomes are log median rent, log median income, and the homeownership rate in percentage points; monetary values are in 2000 dollars before taking logs. Three controls per recipient tract are matched on pre-construction characteristics, then pooled and reweighted. Same-division controls come from the recipient tract's Census division; same-region controls come from its Census region; national controls may come from any metropolitan area in the sample. Donor pools exclude recipient and nearby tracts. Bars show 95% confidence intervals with Conley standard errors (2 km cutoff). The reference period is $t = -1$, the decade before construction.

Figure 11: Spillover Effects: Black Population Share in Nearby Neighborhoods



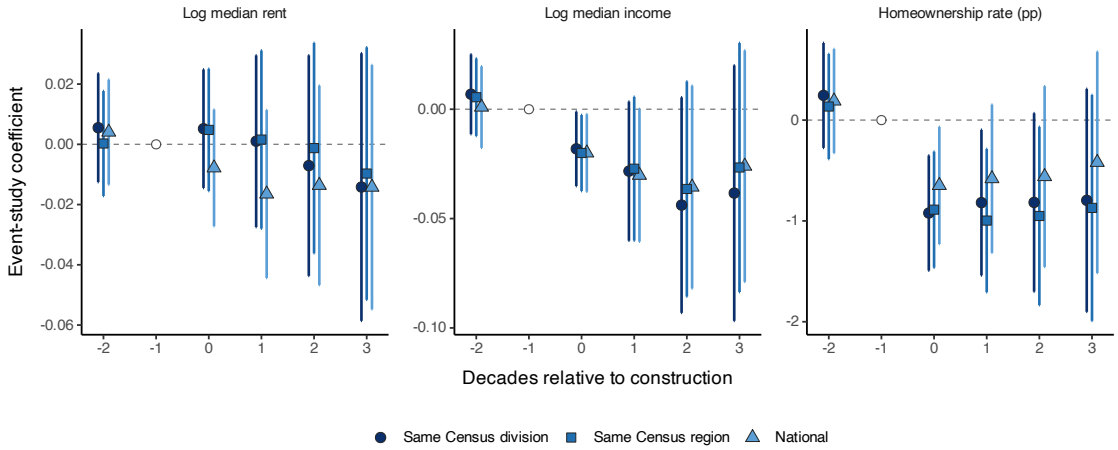
Note: Weighted least-squares event-study estimates of spillover effects on Black population share, in percentage points, using Equation 2. Nearby neighborhoods contain no qualifying public housing project but lie within 1 km of one ($N = 3,503$). Three controls per nearby tract are matched on pre-construction characteristics, then pooled and reweighted. Same-division controls come from the nearby tract's Census division; same-region controls come from its Census region; national controls may come from any metropolitan area in the sample. Donor pools exclude recipient and nearby tracts. Bars show 95% confidence intervals with Conley standard errors (2 km cutoff). The reference period is $t = -1$, the decade before construction.

Figure 12: Spillover Effects: Population in Nearby Neighborhoods, Percentages and Levels



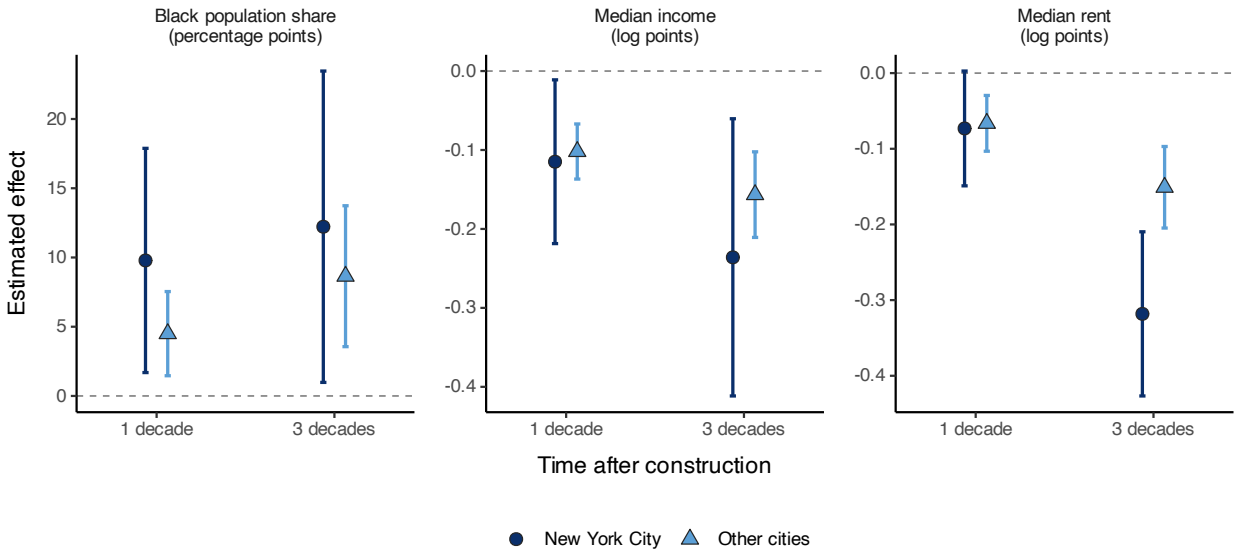
Note: Spillover effects on total, Black, and white population. The top row reports PPML percentage effects, $100[\exp(\beta) - 1]$, from Equation 3; the bottom row reports weighted least-squares effects in residents from Equation 2. Both start from the same matched samples and matching weights. Nearby neighborhoods contain no qualifying public housing project but lie within 1 km of one ($N = 3,503$). Three controls per nearby tract are matched on pre-construction characteristics, then pooled and reweighted. Same-division controls come from the nearby tract's Census division; same-region controls come from its Census region; national controls may come from any metropolitan area in the sample. Donor pools exclude recipient and nearby tracts. Bars show 95% Conley confidence intervals (2 km cutoff); PPML interval endpoints are transformed to percentages. The reference period is $t = -1$, the decade before construction.

Figure 13: Spillover Effects: Economic and Housing Outcomes in Nearby Neighborhoods



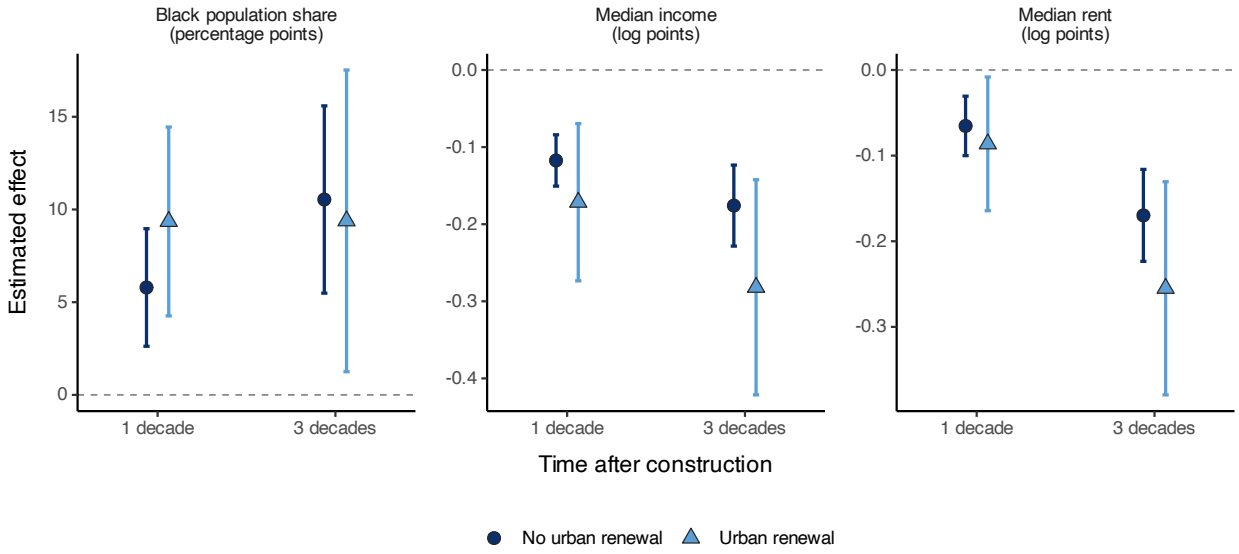
Note: Weighted least-squares event-study estimates of spillover effects, using Equation 2. Outcomes are log median rent, log median income, and the homeownership rate in percentage points; monetary values are in 2000 dollars before taking logs. Nearby neighborhoods contain no qualifying public housing project but lie within 1 km of one ($N = 3,503$). Three controls per nearby tract are matched on pre-construction characteristics, then pooled and reweighted. Same-division controls come from the nearby tract's Census division; same-region controls come from its Census region; national controls may come from any metropolitan area in the sample. Donor pools exclude recipient and nearby tracts. Bars show 95% confidence intervals with Conley standard errors (2 km cutoff). The reference period is $t = -1$, the decade before construction.

Figure 14: New York City vs Other Cities



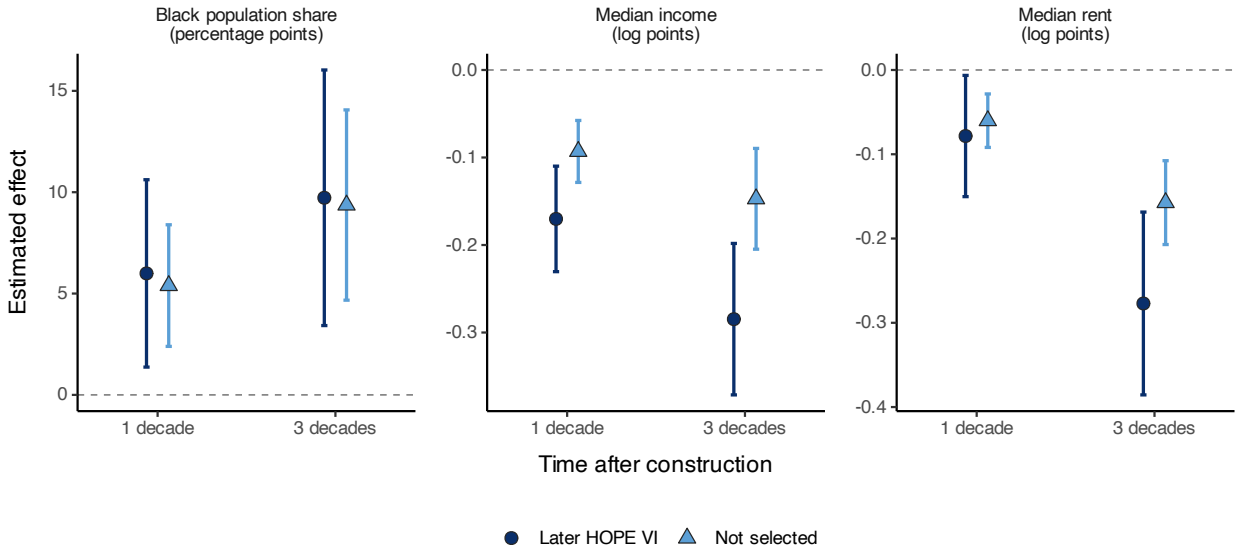
Note: This figure displays heterogeneous treatment effects comparing New York City to other cities in the sample. NYCHA was the largest U.S. housing authority and is often considered better-managed than other PHAs (Bloom 2008). The initial match selects three controls per recipient tract on pre-construction characteristics from eligible tracts across the metropolitan areas in the sample, excluding recipient and nearby tracts. Recipient tracts are compared with selected controls in the same city group (New York City or other cities), retaining matching weights rescaled within group and construction cohort. Estimates follow Equation 2. The three panels show Black population share in percentage points and median income and rent in natural logs. Within each panel, points compare effects one and three decades after construction; colors and shapes distinguish the two subgroups. Bars show 95% confidence intervals with Conley standard errors (2km cutoff). See Section 6.4 for interpretation.

Figure 15: Heterogeneity by Urban Renewal Status



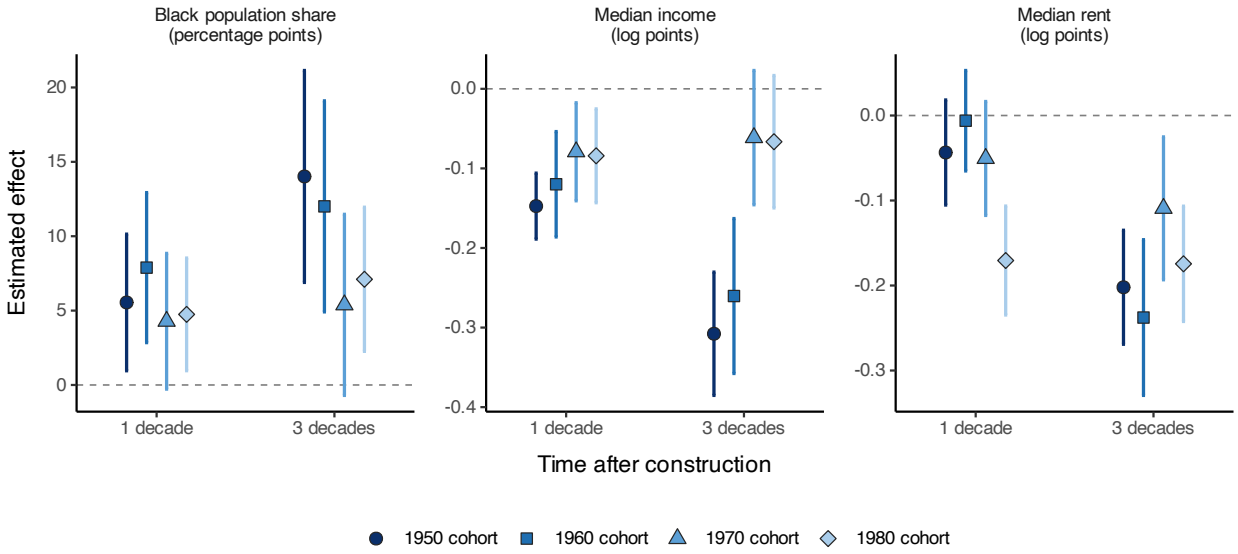
Note: This figure displays heterogeneous treatment effects by urban renewal status, comparing tracts with at least 5% overlap with urban renewal project boundaries to non-urban renewal tracts. This analysis addresses potential confounding between the two programs and tests whether effects differ where both operated. The initial match selects three controls per recipient tract on pre-construction characteristics from eligible tracts across the metropolitan areas in the sample, excluding recipient and nearby tracts. Recipient tracts are compared with selected controls in the same urban-renewal group, retaining matching weights rescaled within group and construction cohort. Estimates follow Equation 2. The three panels show Black population share in percentage points and median income and rent in natural logs. Within each panel, points compare effects one and three decades after construction; colors and shapes distinguish the two subgroups. Bars show 95% confidence intervals with Conley standard errors (2km cutoff). See Section 6.4 for interpretation.

Figure 16: Heterogeneity by Eventual HOPE VI Selection



Note: This figure displays treatment effects by eventual HOPE VI selection, comparing recipient tracts whose projects were later selected for a HOPE VI revitalization grant (195 tracts) with those whose projects were not (983 tracts). Eventual selection is a retrospective marker of project distress, not a treatment, and the comparison is descriptive. The initial match selects three controls per recipient tract on pre-construction characteristics from eligible tracts across the metropolitan areas in the sample, excluding recipient and nearby tracts. Selected controls are pooled and reweighted within construction cohort, and both groups share the same controls. Estimates follow Equation 2. The three panels show Black population share in percentage points and median income and rent in natural logs. Within each panel, points compare effects one and three decades after construction; colors and shapes distinguish the two subgroups. Bars show 95% confidence intervals with Conley standard errors (2km cutoff). See Section 6.4 for interpretation.

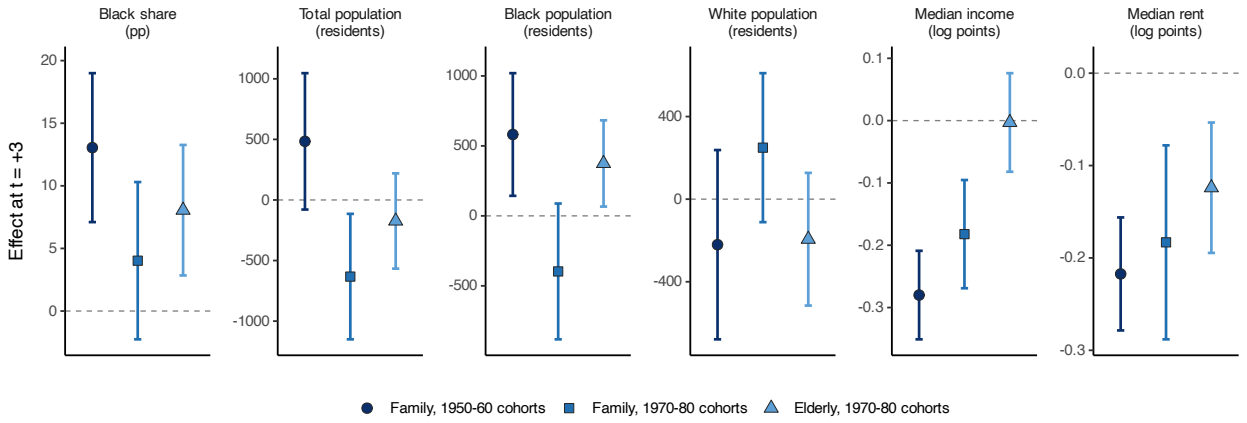
Figure 17: Heterogeneity by Construction Cohort



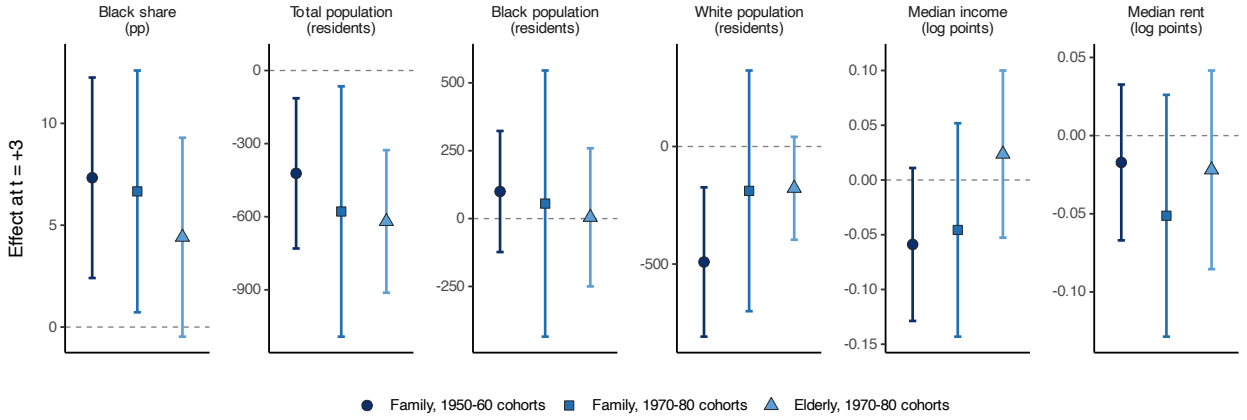
Note: This figure displays separate treatment effects for recipient tracts in each first-treated Census cohort: 1950 (first qualifying project completed April 1940 to March 1950), 1960 (April 1950 to March 1960), 1970 (April 1960 to March 1970), and 1980 (April 1970 to 1973). The initial match selects three controls per recipient tract on pre-construction characteristics from eligible tracts across the metropolitan areas in the sample, excluding recipient and nearby tracts. Each cohort retains its own selected controls and matching weights, and the four cohort paths are estimated jointly with tract-by-cohort and cohort-by-year fixed effects. The three panels show Black population share in percentage points and median income and rent in natural logs. Within each panel, points compare effects one and three decades after construction; colors and shapes distinguish cohorts. Bars show 95% confidence intervals with Conley standard errors (2km cutoff). Cohorts differ in places, projects, and calendar periods. See Section 7.1 for interpretation.

Figure 18: Heterogeneity by Project Type and Years Opened

(a) Recipient neighborhoods



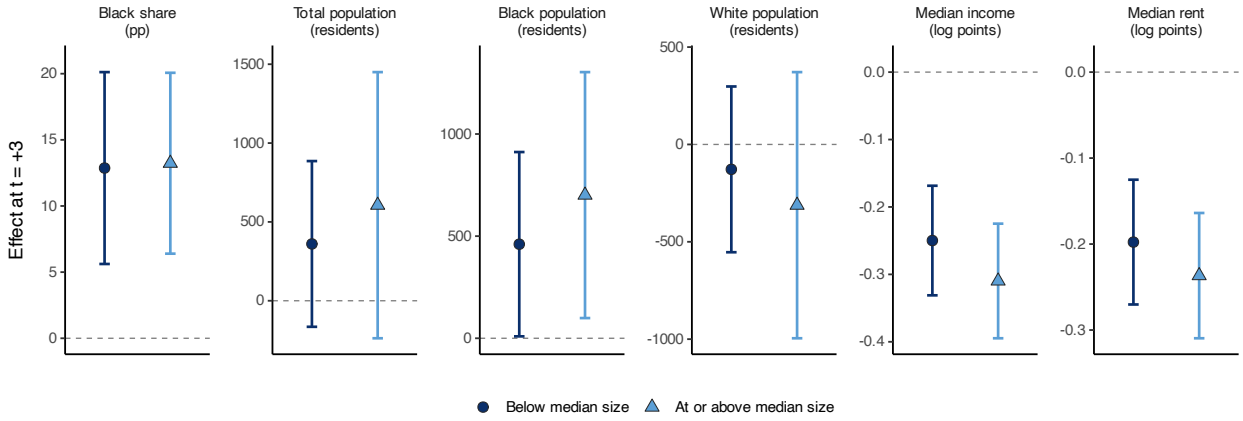
(b) Nearby neighborhoods



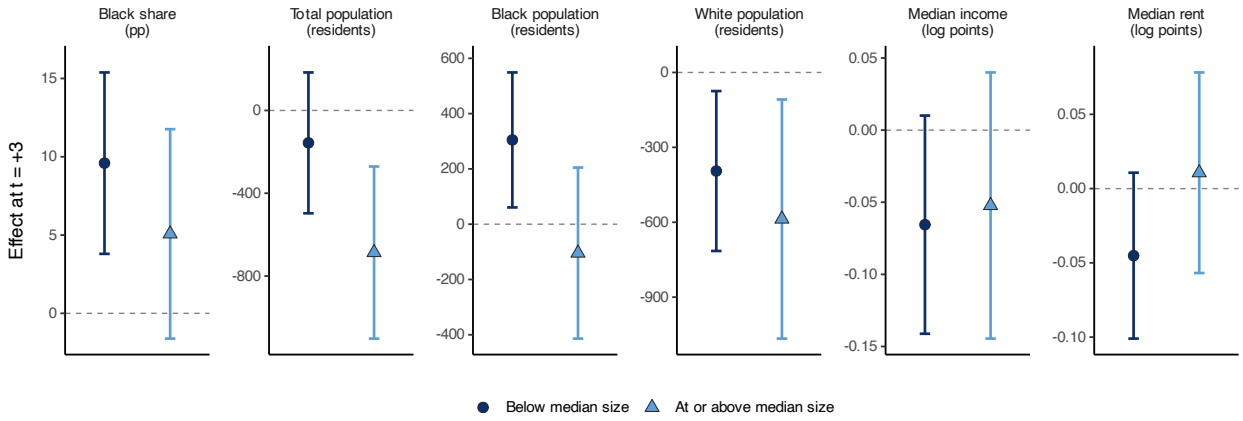
Note: Effects three decades after construction for recipient and nearby neighborhoods by project type and the years projects opened: family projects opened 1940–1960 (1950 and 1960 cohorts), and family and elderly projects opened 1960–1973 (1970 and 1980 cohorts). Recipient counts are 563, 159, and 301; nearby counts are 2,010, 252, and 756. Tracts exposed to both family and elderly projects or to a project with no type on record, and early-cohort tracts exposed only to elderly projects, are left out. Three controls per recipient or nearby tract are initially matched on pre-construction characteristics from national donors; all groups share the selected controls and matching weights of the main design. Panels report weighted least-squares effects: Black share in percentage points, population counts in residents, and median income and rent in log points. Bars show 95% Conley intervals (2 km); the reference decade is $t = -1$.

Figure 19: Early Family Projects by Project Size

(a) Recipient neighborhoods



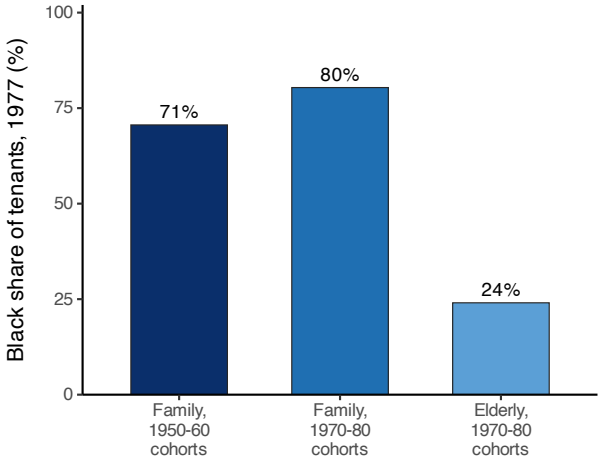
(b) Nearby neighborhoods



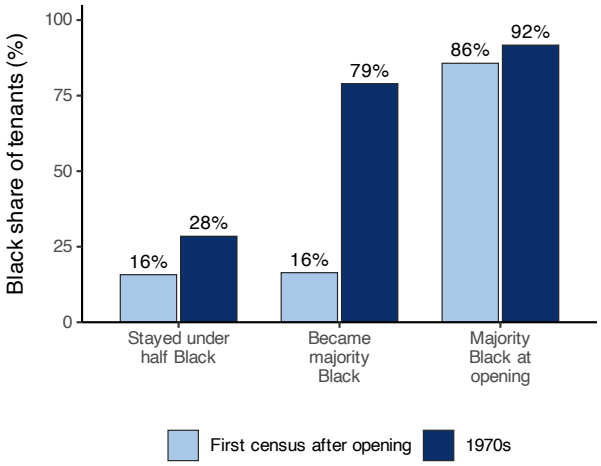
Note: Effects three decades after construction for recipient and nearby neighborhoods of family projects opened 1940–1960, split at the median project size. Recipient tracts are split at 535 units, the median total size of the projects in the tract (counts 280 and 283); nearby tracts at 720 units, the median size of the projects that first expose them (counts 1,003 and 1,007). Three controls per recipient or nearby tract are initially matched on pre-construction characteristics from national donors; both groups share the selected controls and matching weights of the main design. Panels report weighted least-squares effects: Black share in percentage points, population counts in residents, and median income and rent in log points. Bars show 95% Conley intervals (2 km); the reference decade is $t = -1$.

Figure 20: Black Share of Project Tenants

(a) In 1977, by project type and years opened



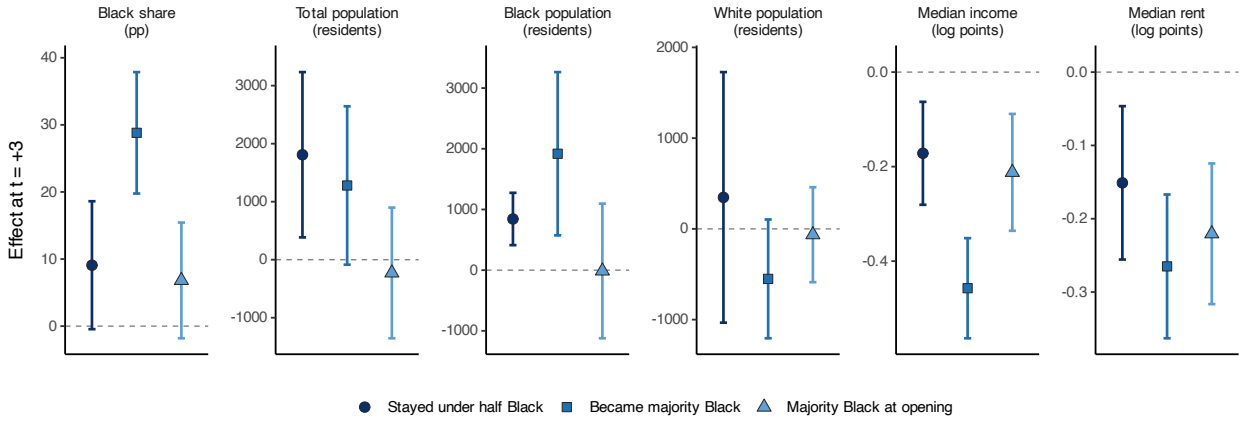
(b) At opening and in the 1970s, by tenant path



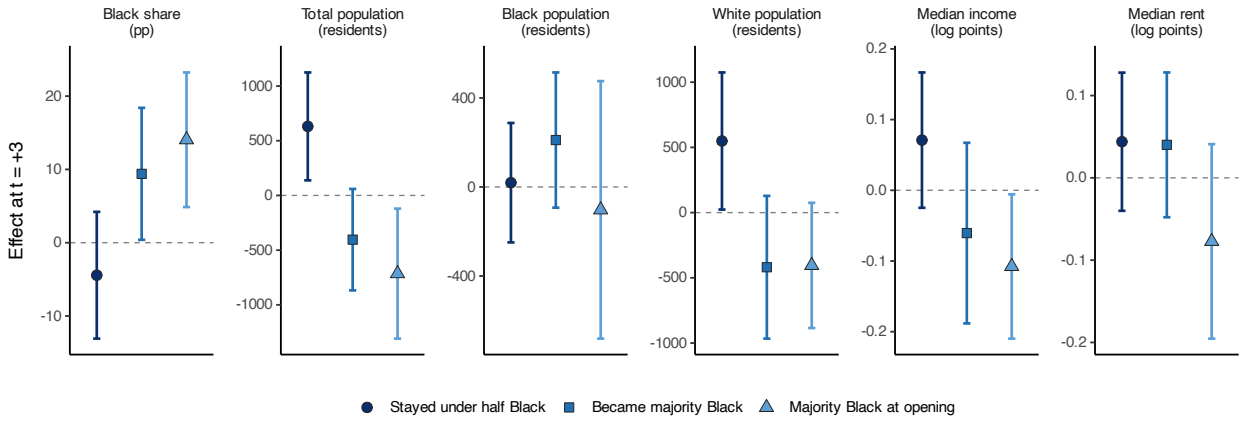
Note: Means over recipient tracts of the Black share of tenant households in the project with the most public housing units in the tract. Panel (a) uses the 1977 Picture of Subsidized Households (PSH) for the recipient tracts of Figure 18 with a 1977 share (458, 139, and 293 tracts). Panel (b) covers recipient tracts of the 1950 and 1960 cohorts whose largest project opened in the tract's cohort decade and has a measured Black share of tenants at its first census (225 tracts, 103 in New York City), grouped by whether that share was below one-half at opening and in the 1970s (82 tracts), below one-half at opening and at least one-half in the 1970s (59), or at least one-half at opening (84). The 1970s share is HUD's 1977 share for federal projects and the December 1971 count for New York City projects.

Figure 21: Heterogeneity by the Racial Path of Project Tenants

(a) Recipient neighborhoods



(b) Nearby neighborhoods



Note: Effects three decades after construction for recipient and nearby neighborhoods of projects opened 1940–1960, grouped by the Black share of the project's tenants at opening and in the 1970s as in Figure 20(b). Recipient counts are 82, 59, and 84; nearby counts are 361, 236, and 361. A nearby tract takes a group when every recipient tract that first exposes it falls in that group. Three controls per recipient or nearby tract are initially matched on pre-construction characteristics from national donors; all groups share the selected controls and matching weights of the main design. Panels report weighted least-squares effects: Black share in percentage points, population counts in residents, and median income and rent in log points. Bars show 95% Conley intervals (2 km); the reference decade is $t = -1$.

Table 7: Long-Run Outcomes in Recipient and Nearby Neighborhoods

	Recipient tracts		Nearby tracts	
1980 controls	Without	With	Without	With
Panel A: Adult household income rank				
Same Census division	-2.45^{***}	-0.19	-1.32^{**}	-0.39
	(0.65)	(0.44)	(0.60)	(0.41)
Same Census region	-2.62^{***}	-0.41	-1.50^{**}	-0.61
	(0.71)	(0.46)	(0.64)	(0.43)
National	-1.88^{***}	-0.07	-1.19^{**}	-0.43
	(0.65)	(0.40)	(0.59)	(0.39)
Panel B: Incarceration rate				
Same Census division	0.84^{***}	0.26	0.34^{*}	0.09
	(0.22)	(0.18)	(0.19)	(0.14)
Same Census region	0.94^{***}	0.36^{*}	0.39^{**}	0.15
	(0.24)	(0.19)	(0.19)	(0.14)
National	0.74^{***}	0.30^{*}	0.38^{**}	0.17
	(0.23)	(0.18)	(0.18)	(0.14)

Notes: Opportunity Atlas outcomes cover children born in 1978–1983, pooling races and genders, with parents at the 25th percentile of national income. Adult household income rank is measured in 2014–2015; incarceration is measured as of April 1, 2010. Income-rank estimates and standard errors are in rank points; incarceration estimates and standard errors are in percentage points. Columns compare recipient tracts and nearby tracts with their matched donors, using each design's existing weights. All specifications include treatment-cohort fixed effects. Standard errors in parentheses allow for spatial correlation following Conley (1999), using a 2-kilometer cutoff and 1950 tract coordinates. 1980 controls are Black population share, median income, and unemployment rate. Adjustment for these post-treatment characteristics is descriptive and does not identify a causal decomposition. $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

A Public Housing Data Construction

Section 4 describes the sources I use to construct the public housing dataset used in this paper. This appendix provides additional details on how I construct the historical project inventory, link later building records to projects, screen project locations, and construct project-level population estimates.

A.1 Linking Modern Building Records to Historical Projects

Most historical building records can be attached to projects using the federal project identifiers described in Section 4. Linking the modern building data from the National Geospatial Data Asset to the historical project inventory requires additional checks, because HUD changed its development identifiers with the transition to Asset Management Projects (AMPs) in 2008. Some historical projects received new codes, and some projects could be grouped under a single AMP. A modern development code need not identify the same project as an earlier HUD code.

I link the NGDA data to the historical project inventory in two stages. First, I identify the historical project associated with each modern development. I compare standardized development names with historical project names from the HUD-951 and the 1997 and 2000 Picture of Subsidized Households files, linked to the CDD. I restrict comparisons to the same housing authority, and use locations to assess candidate matches. For developments that remain unresolved, I use a crosswalk that I constructed by matching names and locations between the 2007 and 2008 Picture of Subsidized Households files within each housing authority.

Second, I check which historical project each building belonged to. This step is necessary because identifying the project associated with an AMP does not establish that all of its buildings belonged to that project. I use individual building names and proximity to historical project locations to identify buildings belonging to other projects. When these checks reveal that an AMP contains buildings from multiple historical projects, I reassign buildings with a supported historical identity and leave ambiguous remaining buildings unmatched.

New York City requires a separate linkage. My historical inventory uses the federal, state, and city project identifiers in the 1973 NYCHA *Project Data Book*, while the NYCHA *Address Guide* identifies buildings by development name. I link these records using standardized names and locations, supplemented by reviewed mappings for known aliases and divided developments. I also require that name matches must be geographically con-

sistent with the historical project location. I allow one historical project to link to multiple address clusters when the address guide lists its sections separately.

A.2 Estimating Public Housing Project Populations

This section describes the methodology for estimating the population of each public housing project using the 1977 Picture of Subsidized Households (PSH) dataset. The 1977 PSH dataset reports household-level demographic information for public housing residents, including the number of households in the project, racial composition, household size, income, and age structure. However, it does not provide direct population counts.

To estimate total, white, and Black populations for each project p, I multiply the number of households for group r by the average household size:

$$\text{Population}_{pr} = \text{Households}_{pr} \times \text{Average Household Size}_{p}.$$

One challenge is that approximately 42% of projects lack household size data. To address this, I impute missing values using two categorical variables that explain substantial variation in household composition: (i) *elderly designation* (all elderly, some elderly, none, or not recorded; projects with a blank designation and those coded not applicable form separate cells) and (ii) *racial composition* (simplified into six categories: all white, all Black, all other race, mixed, no white, and other). For each elderly designation $\times$ racial composition cell, I calculate the mean household size across all projects with non-missing data. Missing household size values are then imputed using the mean of the corresponding cell. This imputation strategy leverages the fact that elderly projects have systematically smaller household sizes, and household size varies by the racial composition of the project.

To validate the imputation strategy, I conducted a hold-out test in which I randomly set 20% of projects with observed household size to missing and then imputed them using the cell means calculated from the remaining 80%. The imputation has an R^2 of 0.46 and is essentially unbiased, with a mean error of -0.01 persons.

A.3 Tenant Racial Composition by Census Year

Because the racial composition of public housing changed substantially after projects opened, I estimate each project's Black share of tenants at each census from 1940 to 1970. I hold each project's resident count at its 1970s level, estimated as described above, and let its racial composition vary by census year.

For each census, I take a project's Black share from the first available of three direct

sources. The first is a count of tenants by race printed within three years of the census. The second is the 1940 and 1950 full-count censuses, in which I match residents to each project's building addresses. I use a match when it finds at least 25 household heads and at least one matched household head for every two units. The third is the official designation of a project as white or Black in a city that formally segregated public housing. I take these designations from housing authority lists in Atlanta, the 1947 hearings of the Joint Committee on Housing for Houston and New Orleans (U.S. Congress, Joint Committee on Housing 1947), court records for Baltimore and Dallas, and a historic preservation filing for Memphis, and I apply each until the city desegregated its projects.[16]

In 1970, projects without a direct source take their 1977 share. In earlier years, I interpolate between a project's nearest known shares or, when no earlier share exists, use its nearest later one. I then shift the imputed shares of federal projects within each city by a single log-odds adjustment weighted by units, so that they match the nearest printed citywide total of Black tenants within three years of the census. I count Hispanic tenants as white, matching the census tract data, where white is a race category that includes Hispanic residents every year. From 1980 on, each project keeps its 1977 racial composition.

A.4 Family and Elderly Designation

I classify each project as housing families, elderly tenants, or both, using the first source that records its designation. The first is the 1977 Picture of Subsidized Households, which records whether a project houses no elderly, some elderly, or only elderly households. The second is the number of elderly units in HUD's 1973 directory, where I treat a blank entry as no elderly units for projects completed by 1965. For projects in neither, I use housing authorities' own project lists, which count family and elderly apartments in Chicago and Boston, flag elderly projects in San Francisco and Washington, and report residents per apartment in New York City, where I classify developments with fewer than two residents per apartment as elderly. For the remaining projects, I use New York State's 1963 *Statistical Summary of Programs* for the state's state-aided projects and, for a few federal projects, the share of household heads aged 62 or older in HUD's 1974 tenant household records. I then classify each recipient and nearby tract by the designations of its projects.

[16] In other Southern cities absent from the Public Housing Administration's 1958 directory of authorities with open occupancy, I treat a census match that is under 5% or over 95% Black in every census matched as a designation through 1960.

Table A1: Public Housing Inventory and Analysis Sample

	Assembled inventory	Included	Excluded
A. Sample coverage			
Number of projects	7,666	1,213	6,453
Share of all inventoried projects (%)	100.0	15.8	84.2
Housing units	980,476	438,003	542,473
Share of all inventoried units (%)	100.0	44.7	55.3
B. Project size			
Mean units per project	127.9	361.1	84.1
Median units per project	60	210	50
Fewer than 50 units (%)	38.9	0.0	46.2
50–199 units (%)	44.4	44.9	44.4
200–499 units (%)	12.3	33.6	8.2
500–999 units (%)	3.0	14.0	0.9
1,000 or more units (%)	1.4	7.4	0.2
C. Completion period			
1935–1940 (%)	2.0	2.2	1.9
1941–1950 (%)	6.8	15.6	5.1
1951–1960 (%)	25.2	24.5	25.4
1961–1970 (%)	46.3	38.2	47.8
1971–1973 (%)	19.8	19.5	19.8
D. Census region			
Northeast (%)	20.0	39.7	16.3
Midwest (%)	20.9	27.9	19.6
South (%)	49.7	20.2	55.3
West (%)	9.4	12.1	8.9

Notes: The assembled inventory combines geolocated projects from the 1973 Consolidated Development Directory with digitized local inventories. It covers completed projects through 1973 and excludes leased housing and Indian housing programs. Included projects are those linked to recipient tracts in the balanced analysis sample, including later construction in already-treated tracts. Excluded projects comprise the remainder of the assembled inventory. Exclusion reflects all project and neighborhood restrictions described in the text. Project-size, completion-period, and regional percentages are shares of projects within each column. Size statistics omit 3 projects with missing unit counts, all excluded from the analysis sample; housing-unit totals sum observed values. A project's full unit count is counted once even if its buildings span several tracts. Regions follow the Census classification, with the District of Columbia in the South.

B Census Tract Data Construction and Harmonization

B.1 Data Sources and Variable Availability

I combine published Census tabulations distributed by NHGIS (Manson et al. 2022) with IPUMS full-count Census microdata for 1930 and 1940 (Ruggles et al. 2022). The published tabulations provide the primary inputs for 1950–2000. For 2010, I use decennial Census counts for population, age, and housing tenure, and 2006–2010 American Community Survey (ACS) estimates for income, rent, home value, employment, and education. The ACS measures therefore describe a five-year period, rather than a single census date.

Published tract coverage is more limited in 1930 and 1940, so I aggregate full-count microdata to enumeration districts, using the 1930 and 1940 boundary shapefiles from the Urban Transition Historical GIS Project (Logan, Minca, et al. 2024). For 1940 districts outside these maps, I use the National Archives' written descriptions of enumeration districts, which name the census tract each district lies in. These descriptions place 41,324 districts in 10,543 tracts. I use a tract's sum only where it reproduces the tract's published 1940 population within 10 percent. I construct the crosswalk to 1950 NHGIS tracts using each district's area share within each tract (Section B.2). These data serve two purposes. First, they add tract-years absent from the published population tabulations. Second, within tract-years already covered by the published data, they fill missing income, rent, home value, labor-force participation, unemployment, age-share, and tenure measures. For these variables, I choose separately for each tract-year: I retain the published-data value when available and otherwise use the full-count measure. Full-count data also provide early income measures absent from the published tract tables.

Table A2 distinguishes the underlying sources from the format of the inputs. Population counts and the counts underlying shares are additive across areas. Income, rent, and home value are represented by grouped frequency distributions; I allocate the counts in each bin to 1950 boundaries before calculating the median, as described in Section B.2.1.

Income definitions change across these sources. In 1930, the Census did not collect income, so I construct a family-level occupational-income proxy by summing, across family members, the machine-learning-adjusted occupational income scores of Saavedra and Twinam (2020). In 1940, I sum family members' reported wage and salary income; this measure excludes nonwage income. The 1950 distribution covers families and unrelated individuals and excludes unreported income. From 1960 onward, I use family-income distributions. Thus, the early income measures differ in both what they capture and the units over which the median is calculated.

Table A2: Sources and Formats of Census Tract Characteristics

	Source				
Characteristic	1930	1940	1950–2000	2010	Input format
Population by race	Both	Both	Census	Census	Counts
Median income	Microdata	Microdata	Census	ACS	Grouped counts
Median rent	Both	Both	Census	ACS	Grouped counts
Median home value	Both	Both	Census	ACS	Grouped counts
Age structure	Microdata	Both	Census	Census	Counts
Homeownership	Microdata	Both	Census	Census	Counts
Labor-market rates	Microdata	Both	Census	ACS	Counts
Educational attainment	—	Both	Census	ACS	Counts

Notes: Census denotes published decennial Census tabulations; Microdata denotes IPUMS full-count records aggregated to tracts. Both indicate that published tabulations and full-count records contribute to the early-year data, with the source-combination rule described in the text. ACS denotes 2006–2010 five-year estimates. Labor-market rates are labor-force participation and unemployment; educational attainment refers to completed schooling, which is unavailable in 1930.

Age shares distinguish residents younger than 18, ages 18–64, and ages 65 and older. Full-count records and published tables for 1970–2010 identify the under-18 population directly. Published tables for 1940–1960 combine ages 15–19, so I approximate the under-18 population by adding three-fifths of that category to the population younger than 15, assuming equal population at each age within the category.

B.2 Tract Boundary Harmonization

I allocate each source tract or enumeration district to 1950 tracts in proportion to its area of overlap. Before calculating areas, I combine all polygon records belonging to the same source unit. For enumeration districts, I identify the source unit by state, city, census year, and enumeration-district code. This ensures that a district represented by several polygons contributes its count only once. I measure both source areas and overlaps in a common equal-area projection. Figure A1 illustrates the resulting weights for two Chicago examples.

Let A_i denote the full area of source unit i, before restricting it to the 1950 tract map, and let a_{ij} denote its overlap with 1950 tract j. The allocation weight is

$$w_{ij} = \frac{a_{ij}}{A_i}.$$

If the 1950 tract map covers only part of a source unit, its weights sum to less than one. I allocate only the share of its count corresponding to the covered area; the share outside

the map remains unallocated. For example, a source tract with 40% of its area inside the 1950 map contributes 40% of its count. Source units with no overlap contribute nothing.

For count variables such as population, housing units, and employment, I sum contributions from the source units overlapping each 1950 tract:

$$X_j = \sum_{i \in I_j} w_{ij} \cdot X_i$$

where X_i is the count in source unit i and I_j is the set of source units overlapping 1950 tract j. For rates and shares, I allocate the numerator and denominator counts separately and calculate their ratio after aggregation. I apply the same weights to the counts in each income, rent, or home-value category before calculating medians, as described below. This method assumes that the counted population or housing units are uniformly distributed within each source unit, which may introduce measurement error in areas where they are spatially concentrated. For instance, if a source tract contains a spatially concentrated Black population but overlaps evenly with multiple 1950 tracts, the method would allocate residents uniformly across those tracts, potentially attenuating spatial estimates of segregation.

B.2.1 Calculating Medians from Categorical Distributions

I construct medians after harmonizing the underlying frequency distributions to 1950 tract boundaries (Section B.2). For each income, rent, or home-value bin, I allocate source-area counts using the geographic crosswalk weights and sum the allocated counts within each destination tract. I then calculate the median from this tract-level distribution, rather than averaging the medians of the source areas.

Let n_i be the count in bin i, with bounds $[L_i, U_i)$, and let $N = \sum_i n_i$. The median bin m is the first bin for which $\sum_{i \le m} n_i \ge N/2$. I linearly interpolate within that bin:

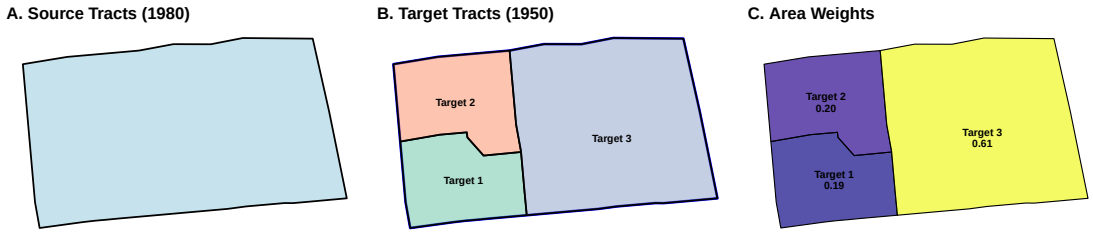
$$\widehat{\text{Median}} = L_m + \frac{N/2 - \sum_{i<m} n_i}{n_m}(U_m - L_m).$$

This calculation assumes a uniform distribution within the median bin. If the median lies in an open-ended top category, I use that category's lower bound. Categories without a monetary value, such as income not reported or no cash rent, are excluded from the calculation and its denominator. For income, the relevant units follow the definitions above; for rent and home value, they are the housing units represented in the respective distribution.

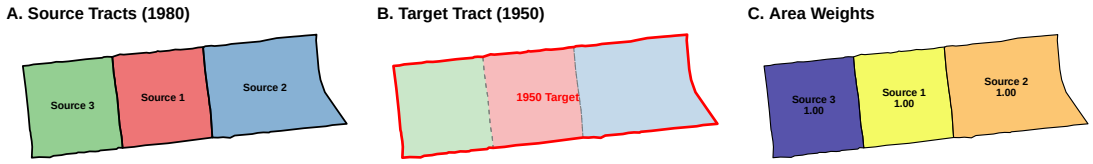
After harmonization and median construction, I convert monetary values to 2000 dol-

Figure A1: Census Tract Harmonization Examples

(a) Split Pattern: One 1980 tract → Multiple 1950 tracts



(b) Merge Pattern: Multiple 1980 tracts → One 1950 tract



Note: These figures demonstrate the area-reweighting methodology for harmonizing 1980 census tracts to 1950 boundaries in a particular example in Chicago. Panel (a) shows a split pattern where one 1980 source tract is divided across multiple 1950 target tracts, with weights representing the proportion of area allocated to each target. Panel (b) shows a merge pattern where multiple 1980 source tracts contribute to a single 1950 target tract.

lars using the Consumer Price Index from Officer and Williamson (2025). For the 1930 occupational-income proxy, which is expressed in 1950 dollars, I apply the 1950-to-2000 deflator.

B.3 Opportunity Atlas Outcomes

I harmonize Opportunity Atlas outcomes from 2010 to 1950 tract boundaries using the same source-area overlap shares. Within each destination tract, I average the source-tract outcomes, weighting by the overlap share multiplied by the Atlas count of children with parental income below the median. I evaluate outcomes at parental income percentile 25; this evaluation point differs from the below-median population used for weighting. I re-

tain the supplied outcome estimates, including negative fitted incarceration rates. These crosswalk weights construct the tract outcomes; the neighborhood comparisons then apply the matched-sample weights described in Section 8.

C Supplementary Results and Robustness Checks

This appendix reports supplementary results and sensitivity checks for the site-selection and neighborhood analyses.

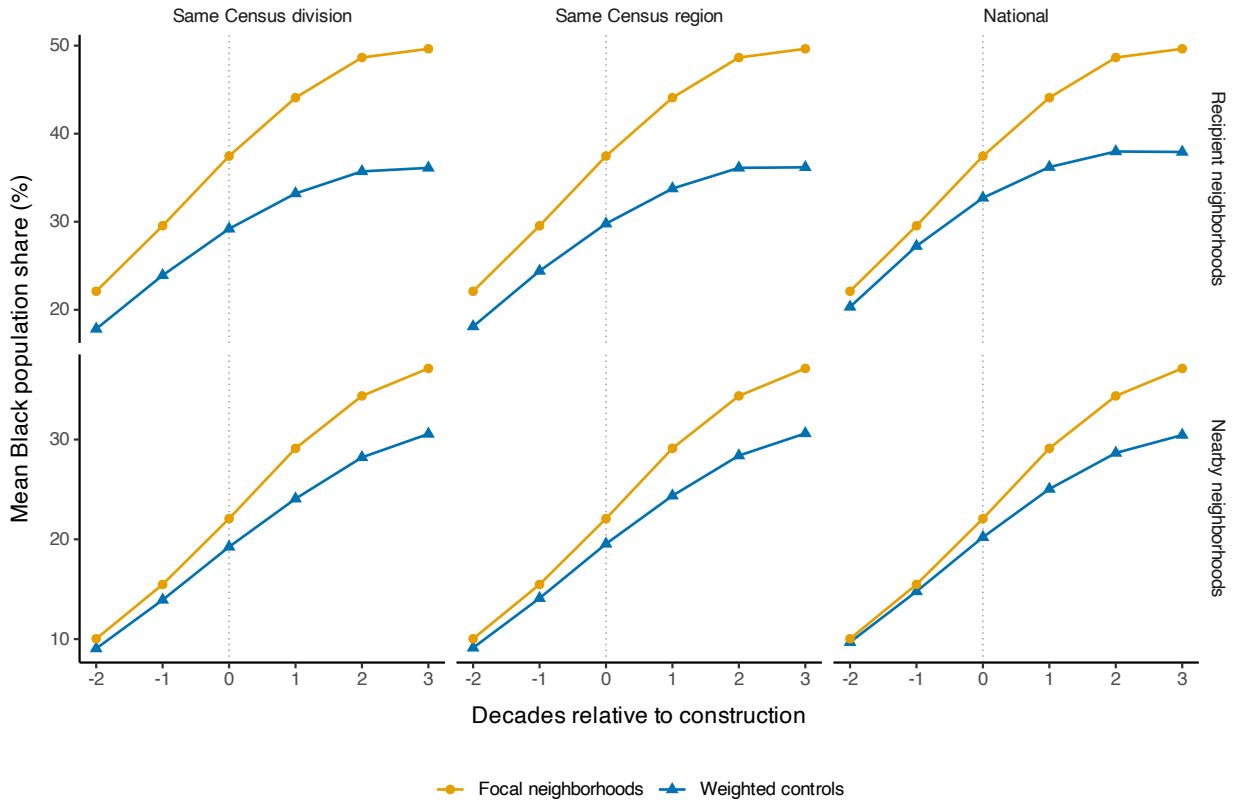
C.1 Chicago Site Selection

Table A3: 1940 Neighborhood Characteristics: Proposed vs Actual Public Housing Locations in Chicago

Characteristic	All tracts	Built sites	Proposed-only sites	Absolute SMD: built vs proposed
Black population share	0.08	0.30	0.07	0.701**
	(0.25)	(0.42)	(0.21)	
Total population (asinh)	8.12	8.47	8.32	0.097
	(1.93)	(1.59)	(1.46)	
Median family income (log)	2.96	2.67	2.88	0.678***
	(0.32)	(0.33)	(0.29)	
Unemployment rate	0.12	0.16	0.14	0.339
	(0.07)	(0.07)	(0.07)	
Median contract rent (log)	5.81	5.61	5.72	0.258
	(0.43)	(0.40)	(0.41)	
Distance from CBD (asinh)	9.33	8.89	9.44	0.514*
	(1.43)	(1.37)	(0.66)	
Redlined (HOLC grade D)	0.25	0.46	0.35	0.230
	(0.43)	(0.50)	(0.49)	
Urban renewal area	0.14	0.25	0.13	0.305
	(0.35)	(0.44)	(0.34)	
Tracts	973	104	23	

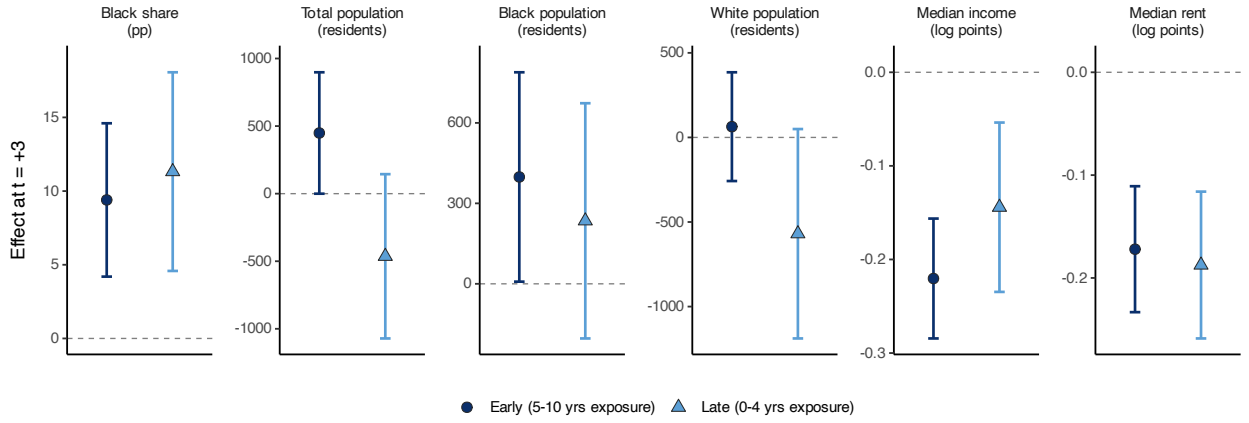
Notes: This table reports 1940 characteristics of Chicago census tracts on 1950 boundaries, in the layout of Table 4. The citywide column includes all tracts in the descriptive sample, including built-site and proposed-only tracts, without the paper's longitudinal sample restrictions. Proposed-only tracts contain sites that the Chicago Housing Authority proposed for family public housing between 1950 and 1966 and did not build, as listed in the authority's 1968 affidavit in *Gautreaux v. Chicago Housing Authority*, and no built public housing sites. The first three columns report tract means, with standard deviations in parentheses beneath them, and the final row reports tract counts. Statistics use nonmissing observations for each characteristic. Income and rent are in natural logs, and population and distance from the central business district (CBD) use the inverse hyperbolic sine (asinh). The final column reports absolute standardized mean differences between built and proposed-only tracts. Stars indicate tests of equality of means for those two groups: *p<0.1, **p<0.05, ***p<0.01.

Figure A2: Black Population Share: Pooled Samples and Weighted Controls Over Event Time



Note: Mean Black population shares for recipient neighborhoods (top row) and nearby neighborhoods (bottom row), with their pooled controls. Controls are matched directly to each recipient or nearby tract from eligible donors in its Census division (same-division), its Census region (same-region), or any metropolitan area in the sample (national). Means use the matching weights, held fixed across event time; lines connect descriptive means rather than event-study coefficients. Each panel uses the corresponding pooled sample. These paths do not establish the absence of spillovers to controls.

Figure A3: Within-Decade Timing: Early vs Late Openers (Pooled)



Note: Estimated effects three decades after construction ($t = 3$) from Equation 2, with three controls per recipient tract selected from eligible donor tracts across the metropolitan areas in the sample (the national donor pool). Estimates are reported separately for early openers (projects opening 5 to 10 years before the next census, 645 recipient tracts) and late openers (0 to 4 years, 297 recipient tracts). The split pools the 1950, 1960, and 1970 cohorts (942 of 1,186 recipient tracts). The 1980 cohort (244 tracts) is excluded because the project inventory ends in 1973, so every project in that cohort opened 7 to 10 years before the 1980 census and the cohort has no late openers. The six panels show Black population share in percentage points; total, Black, and white population in numbers of residents; and median income and rent in natural logs. 95% confidence intervals based on Conley standard errors (2km cutoff). Reference period: $t = -1$.

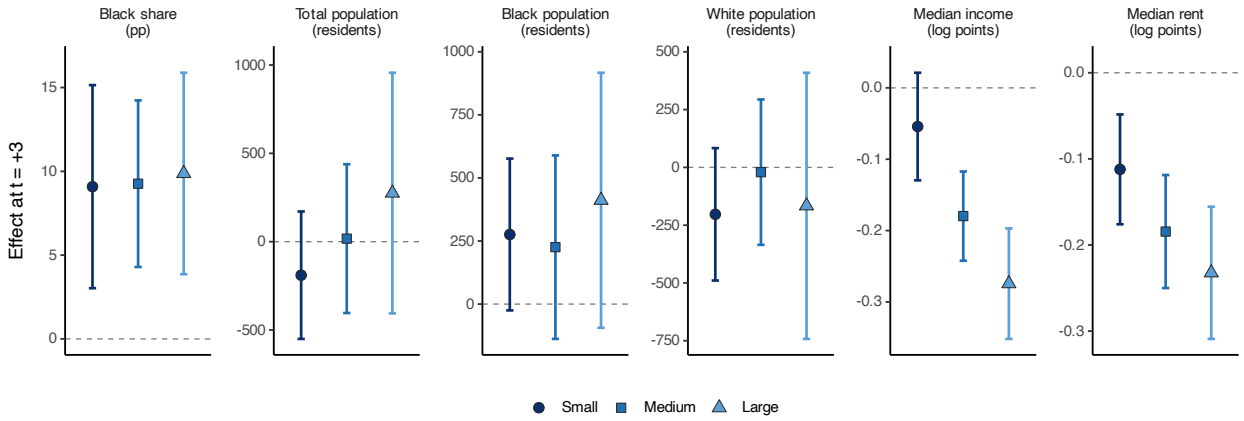
C.2 Project Size in the Full Sample

Larger projects produced larger socioeconomic declines but not larger racial change. I divide recipient neighborhoods into terciles of the number of public housing units built in the tract: fewer than 199 units ($N = 395$), 199–501 units ($N = 394$), and more than 501 units ($N = 397$). Nearby neighborhoods are grouped by terciles of the units of the projects near them. Because project size is a characteristic of recipient neighborhoods rather than of donors, the three groups share the matched controls and weights of the main design. Figure A4 shows the estimates three decades after construction with national donors. The Black-share increase is similar across the three groups, at 9.1 and 9.9 percentage points, and none of the differences is statistically significant under any donor pool. The economic effects are not similar. Median income falls by 24.0% in neighborhoods with the largest

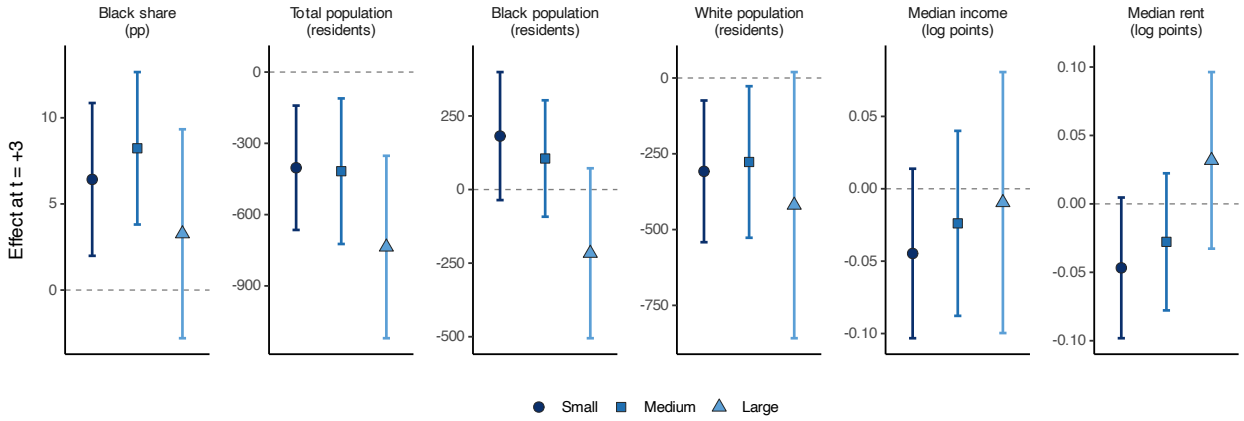
projects, against 5.3% and 16.5% in the other two groups, and the contrast between the largest group and each of the others is significant under every donor pool; rents fall by 20.7% in the largest group. White population does not line up with size in the same way: it falls by 18.1% in the largest group, an estimate that is imprecise with national donors and significant with same-region and same-division donors, and by 203 residents in the smallest group. Nearby neighborhoods around the largest projects show a 32.7% decline in Black population that differs significantly from the other two groups, and a smaller Black-share increase (3.3 percentage points, against 6.4 around the smallest projects), so the largest projects appear to have concentrated Black residents inside the recipient tract rather than in the surrounding neighborhoods.

Figure A4: Heterogeneity by Project Size

(a) Recipient neighborhoods



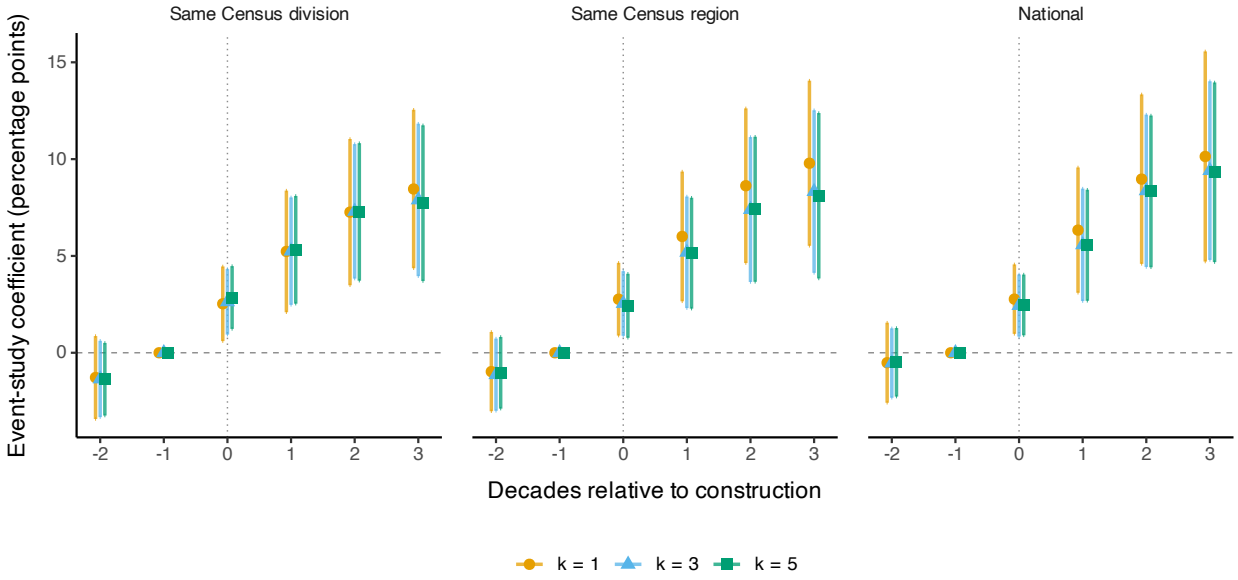
(b) Nearby neighborhoods



Note: Effects three decades after construction for recipient and nearby neighborhoods by project size. Recipient tracts are grouped into terciles of the total number of public housing units built in the tract (cutoffs at 199 and 501 units; counts 395, 394, and 397), and nearby tracts into terciles of the units of the projects near them (counts 1,168, 1,167, and 1,168). Three controls per recipient or nearby tract are initially matched on pre-construction characteristics from national donors; recipient and nearby tracts are excluded from donor pools. All size groups share the selected controls and matching weights of the main design, rescaled within construction cohort. Panels report weighted least-squares effects: Black share in percentage points, population counts in residents, and median income and rent in log points. Bars show 95% Conley intervals (2 km); the reference decade is $t = -1$.

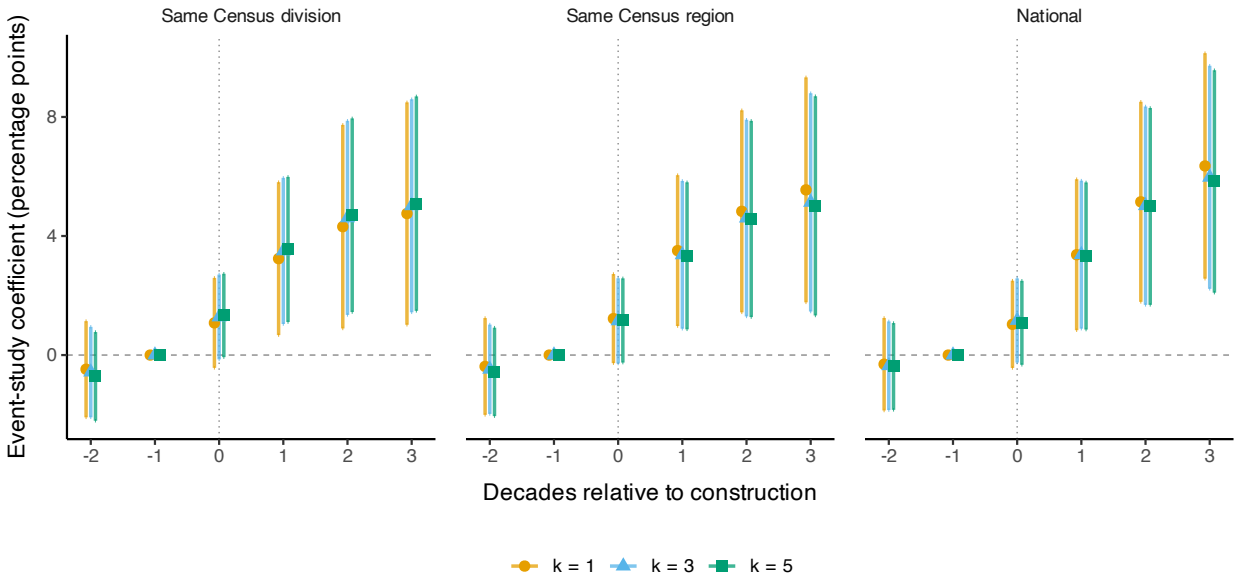
C.3 Sensitivity to the Number of Matched Controls

Figure A5: Black Population Share in Recipient Neighborhoods: $k = 1$, 3, and 5 Matched Controls by Donor Pool



Note: Event-study coefficients from Equation 2 for the Black population share of recipient tracts, in percentage points, re-estimating the main design with one, three, or five matched controls per recipient tract (Section 6.1.1). Panels distinguish controls selected from eligible donors in each recipient tract's Census division (same-division), its Census region (same-region), or any metropolitan area in the sample (national). Bars are 95% confidence intervals with Conley standard errors (2 km cutoff); the reference decade is $t = -1$.

Figure A6: Black Population Share in Nearby Neighborhoods: $k = 1$, 3, and 5 Matched Controls by Donor Pool



Note: Event-study coefficients from Equation 2 for the Black population share of nearby tracts, in percentage points, re-estimating the directly matched nearby design with one, three, or five matched controls per nearby tract (Section 6.1). Panels distinguish controls selected from eligible donors in each nearby tract's Census division (same-division), its Census region (same-region), or any metropolitan area in the sample (national). Bars are 95% confidence intervals with Conley standard errors (2 km cutoff); the reference decade is $t = -1$.

C.4 Spatial Inference

Table A4: Sensitivity to the Conley Distance Cutoff

	Division			Region			National		
		Conley SE			Conley SE			Conley SE	
Outcome	Est.	2 km	5 km	Est.	2 km	5 km	Est.	2 km	5 km
A. Recipient neighborhoods									
Total population (PPML)	0.057	0.045	0.051	0.060	0.047	0.055	0.033	0.044	0.053
Black population (PPML)	0.085	0.113	0.127	0.109	0.122	0.141	0.148	0.114	0.132
White population (PPML)	-0.094	0.063	0.072	-0.061	0.070	0.080	-0.049	0.063	0.074
Black population share (pp)	7.89	2.01***	2.29***	8.32	2.14***	2.41***	9.41	2.35***	2.98***
Median income (log)	-0.186	0.031***	0.025***	-0.168	0.032***	0.027***	-0.170	0.028***	0.025***
Median rent (log)	-0.193	0.030***	0.035***	-0.193	0.030***	0.036***	-0.176	0.027***	0.033***
Homeownership rate (pp)	-6.11	0.87***	0.93***	-6.19	0.84***	0.87***	-5.83	0.83***	0.95***
B. Nearby neighborhoods									
Total population (PPML)	-0.090	0.032***	0.034***	-0.088	0.032***	0.035**	-0.134	0.033***	0.038***
Black population (PPML)	-0.154	0.119	0.135	-0.131	0.119	0.137	-0.073	0.119	0.143
White population (PPML)	-0.153	0.050***	0.051***	-0.145	0.051***	0.051***	-0.154	0.049***	0.053***
Black population share (pp)	5.02	1.83***	2.00**	5.13	1.87***	2.05**	5.97	1.91***	2.38**
Median income (log)	-0.038	0.030	0.031	-0.027	0.029	0.031	-0.026	0.027	0.030
Median rent (log)	-0.014	0.023	0.026	-0.010	0.021	0.025	-0.014	0.021	0.025
Homeownership rate (pp)	-0.80	0.56	0.51	-0.87	0.57	0.51*	-0.42	0.56	0.57

Notes: Effects three decades after construction ($t = 3$), relative to $t = -1$, using three matched controls and the indicated donor pool. Each estimate has Conley standard errors (SE) at 2 km and 5 km; samples, matching weights, and fixed effects are identical. Population rows report PPML log-link coefficients; income and rent are in logs, and shares are in percentage points (pp). Nearby neighborhoods use directly matched controls. Stars beside each SE test whether the coefficient differs from zero using that cutoff: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

C.5 Building-Located Projects

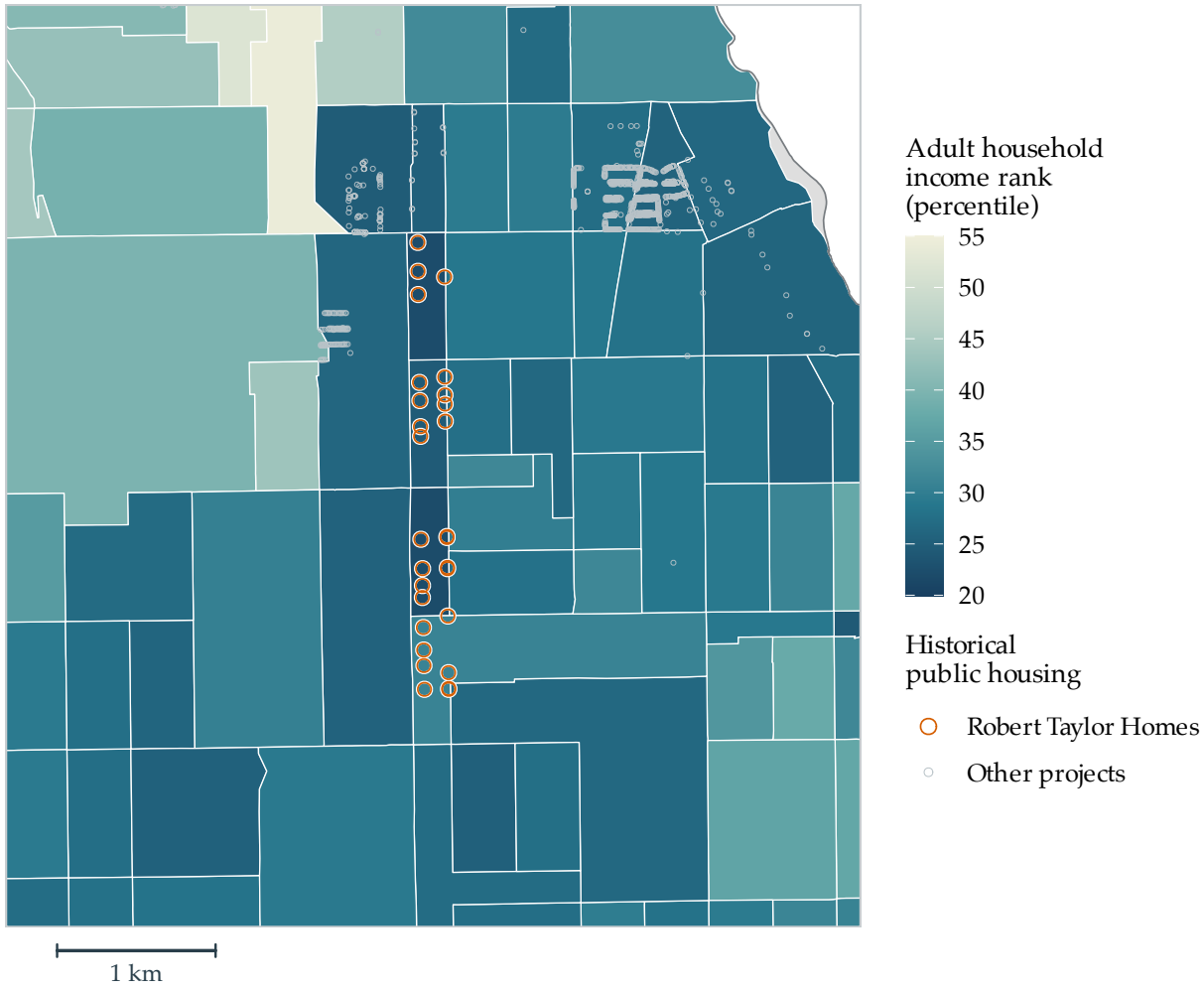
Table A5: Recipient and Nearby Tracts Located by Building Addresses

	Division		Region		National	
Outcome	All	Buildings	All	Buildings	All	Buildings
A. Recipient neighborhoods						
Total population (PPML)	0.057	0.070	0.060	0.080*	0.033	0.075
	(0.045)	(0.046)	(0.047)	(0.047)	(0.044)	(0.049)
Black population (PPML)	0.085	0.107	0.109	0.111	0.148	0.234*
	(0.113)	(0.113)	(0.122)	(0.118)	(0.114)	(0.122)
White population (PPML)	-0.094	-0.100	-0.061	-0.057	-0.049	-0.066
	(0.063)	(0.068)	(0.070)	(0.071)	(0.063)	(0.069)
Black population share (pp)	7.89***	7.28***	8.32***	7.84***	9.41***	9.83***
	(2.01)	(2.10)	(2.14)	(2.18)	(2.35)	(2.55)
Median income (log)	-0.186***	-0.214***	-0.168***	-0.195***	-0.170***	-0.195***
	(0.031)	(0.032)	(0.032)	(0.032)	(0.028)	(0.030)
Median rent (log)	-0.193***	-0.216***	-0.193***	-0.216***	-0.176***	-0.193***
	(0.030)	(0.030)	(0.030)	(0.030)	(0.027)	(0.026)
Homeownership rate (pp)	-6.11***	-6.69***	-6.19***	-6.80***	-5.83***	-6.90***
	(0.87)	(0.95)	(0.84)	(0.92)	(0.83)	(0.91)
Recipient tracts	1,186	967	1,186	967	1,186	967
B. Nearby neighborhoods						
Total population (PPML)	-0.090***	-0.112***	-0.088***	-0.108***	-0.134***	-0.125***
	(0.032)	(0.033)	(0.032)	(0.033)	(0.033)	(0.032)
Black population (PPML)	-0.154	-0.136	-0.131	-0.119	-0.073	-0.087
	(0.119)	(0.126)	(0.119)	(0.125)	(0.119)	(0.126)
White population (PPML)	-0.153***	-0.206***	-0.145***	-0.190***	-0.154***	-0.166***
	(0.050)	(0.053)	(0.051)	(0.054)	(0.049)	(0.053)
Black population share (pp)	5.02***	5.25***	5.13***	5.14***	5.97***	5.03**
	(1.83)	(1.89)	(1.87)	(1.93)	(1.91)	(2.01)
Median income (log)	-0.038	-0.053*	-0.027	-0.043	-0.026	-0.041
	(0.030)	(0.031)	(0.029)	(0.031)	(0.027)	(0.028)
Median rent (log)	-0.014	-0.022	-0.010	-0.020	-0.014	-0.024
	(0.023)	(0.022)	(0.021)	(0.021)	(0.021)	(0.021)
Homeownership rate (pp)	-0.80	-1.09*	-0.87	-1.11*	-0.42	-0.78
	(0.56)	(0.57)	(0.57)	(0.57)	(0.56)	(0.57)
Nearby tracts	3,503	3,057	3,503	3,057	3,503	3,057

Notes: Effects three decades after construction ($t = 3$), relative to $t = -1$, using three matched controls and the indicated donor pool. *All* uses every recipient and nearby tract. *Buildings* keeps recipient tracts in which at least one qualifying project completed in the tract's first treatment decade is located from geocoded building addresses, and nearby tracts first placed within 1 km of such a project; tracts whose status rests only on a single project-level coordinate leave the sample, and controls are re-selected from the same donor pool. Population rows report PPML log-link coefficients; income and rent are in logs, and shares are in percentage points (pp). Conley standard errors (2 km) in parentheses: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

C.6 Neighborhood Opportunity

Figure A7: Upward Mobility around Robert Taylor Homes



Note: The map shows upward mobility in 2010 census tracts around Robert Taylor Homes in Chicago. Shading gives expected adult household income rank in the national distribution for children born in 1978–1983 whose parents were at the 25th income percentile, pooling races and genders. Income is measured in 2014–2015. Darker tracts have lower mobility; the color scale spans 20–55. Orange circles mark the 28 Robert Taylor building/address locations used to assign treatment. Gray circles mark other projects, using building/address locations where available and project coordinates otherwise. Markers have fixed sizes and do not represent building footprints. The map describes neighborhood outcomes rather than effects of public housing. Sources: Opportunity Atlas (Chetty, Friedman, et al. 2018), NHGIS tract boundaries, and the historical housing data described in Section 4.